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Private University Estd. in Karnataka State by Act No. 41 of 2013

Itgalpura, Rajankunte, Yelahanka, Bengaluru – 560064



USE OF DIGITAL TECHNOLOGY TO CALCULATE WATER FOOTPRINTS FOR DIFFERENT DAILY USE ITEMS

A PROJECT REPORT

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BACHELOR OF TECHNOLOGY

IN

INFORMATION SCIENCE AND TECHNOLOGY

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PRESIDENCY SCHOOL OF INFORMATION SCIENCE AND TECHNOLOGY

BONAFIDE CERTIFICATE

Certified that this report “ USE OF DIGITAL TECHNOLOGY TO CALCULATE WATER FOOTPRINTS FOR DIFFERENT DAILY USE ITEMS” is a Bonafide work of “BOLLA SAGARIKA (20221IST0017), GANDAVARAM SRAVANTHI (20221IST0002), BATHINA MOUNIKA (20221IST0003)”, who have successfully carried out the project work and submitted the report for partial fulfilment of the requirements for the award of the degree of BACHELOR OF TECHNOLOGY in INFORMATION SCIENCE AND TECHNOLOGY during 2025-26.

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DECLARATION

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Abstract

Water is one of the most important resources for sustainable development and human survival. However, because of rapid urbanisation, increasing population, and consumption patterns, freshwater availability is under extreme pressure. The recognition and keeping an eye on the water footprint everyday commodities like food, clothing, and other household goods become highly relevant to promoting responsible usage. This project looks into the real-time computation and analysis of water footprints using digital technologies. The methodology adopts the V-Model framework to ensure structured development, from requirement analysis to testing and evaluation. The project not only highlights the role of technology in advancing sustainable practices but also aligns with global goals such as SDG 6 (Clean Water and Sanitation) and SDG12 (Responsible Consumption and Production). Ultimately, the system provides individuals and communities with actionable insights, encouraging informed decisions and contributing to long-term water conservation.

Besides this, the project assesses the role played by digital technologies in reducing industries' water footprints. Most of the industries are now into the installation of IoT-based water flow sensors, AI-driven recycling systems, and automated treatment plants that monitor water usage across different manufacturing processes. Such innovations reduce operational costs, minimize pollution, and help an organization contribute towards corporate sustainability goals.

The electronic publishing of water footprint for various products from different businesses empowers the consumer to make informed decisions in a rapid fashion. Government and environmental agencies have begun to use electronic tools in ways such as establishing regulations concerning water use, tracking the condition of river basins and conducting conservation programs.

The conclusion of the report stated that despite Advanced Technology (Digital Technology)DT, relating to the Assessment of Water Footprint, having been recognised as a first-generation technical tool (i.e. ASAP) and its independent utility, it is arguably the greatest opportunity for enabling organisations and individuals to develop and improve their commitment to Sustainability.

According to the findings from this report, Digital Technologies assist end-users with carrying out complicated and detail-oriented calculations as well as developing an overall understanding of the impact of water and DT use on the environment, providing factual support to assist in establishing governmental policy, and supplying consumers with data upon which they can base their decisions to ultimately reduce their water consumption and achieve an overall lower carbon footprint.

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ABBREVIATIONS

Abbreviations	Full Form
IoT	Internet of Things
AI	Artificial Intelligence
GIS	Geographic Information Systems
WFN	Water Footprint Network
ML	Machine Learning
SDGs	Sustainable Development Goals
LCA	Life Cycle Assessment
WF	Water Footprint
BI	Business Intelligence
ERP	Enterprise Resource Planning
SCM	Supply Chain Management
RBAC	Role-Based Access Control
NFRs	Non-Functional Requirements
CPUs	Cloud Computing resources
NGO	Non-Governmental Organizations
GDPR	General Data Protection Regulation
CCPA	California Consumer Privacy Act
ESG	Environmental, Social, and Governance
PWA	progressive web application
BW	Blue Water Footprint
GW	Green Water Footprint
GrW	Grey Water Footprint
TWF	Total Water Footprint
UA	User Acceptance Testing

Chapter 1

Introduction

1.1 Background

Water is perhaps the most important resource on the planet because it supports human life through many means. Water is important in agriculture, for industry and is used to maintain the ecological balance. Urbanisation, industrialisation and increasing populations have greatly increased the amount of freshwater that people use, which has resulted in unsustainable patterns of use. A water footprint denotes the amount of fresh water used both in direct and indirect ways to produce goods or services. This is enabled by emerging digital technologies. Besides, knowledge of the water footprint of things used every day-such as food, clothes, and energy-enables people and companies to make more informed decisions about resource use in ways that are more sustainable.

When you calculate your Water Footprint, it lets you know how much water is being consumed in your household (i.e., through cooking and drinking) as well as the amount of water you consume through products you purchase and use daily. An example would be the amount of water used to create; One kg of rice will need approximately 2,500 litres of water to produce, a cotton T-shirt needs approximately 2,700 litres of water to produce, and one cup of coffee takes about 140 litres of water to make. Each of these products illustrates how they contribute to the overall consumption of water for our daily products.

It encompasses both freshwater used in production processes as well as the amount required to sustain the resource recovery process. The concept encompasses three main parts which are green water (rainwater used in the production process), blue water (surface and groundwater) and grey water (freshwater needed for the dilution of the pollutants to a safe level). With this approach, one can see the "virtual" or hidden water that is present in daily consumption items like food, clothes, and household goods. For instance, the water consumption for the production of 1 kilogram of rice can exceed more than 2,500 Liters, while the water consumption for a single cotton shirt can be as high as over 2,700 Liters. These data points are a revelation of the extent to which consumer behaviour and industrial practices play a role in causing water scarcity around the world.

Mobile apps, web-based water footprint calculators, cloud-stored databases, IoT, AI, GIS, and remote sensing are among the technologies that are capable to deliver precise and immediate information about water use. The use of such technologies leads to the simplification of difficult calculations, reduction of human mistakes, and giving light to water footprint assessment coming from the students, researchers, policymakers, and people in general. For instance, IoT devices monitor the water consumption in domestic and industrial sectors, future water requirements are estimated by AI models, and the water footprint of a large number of products is provided by the Water Footprint Network using global datasets.

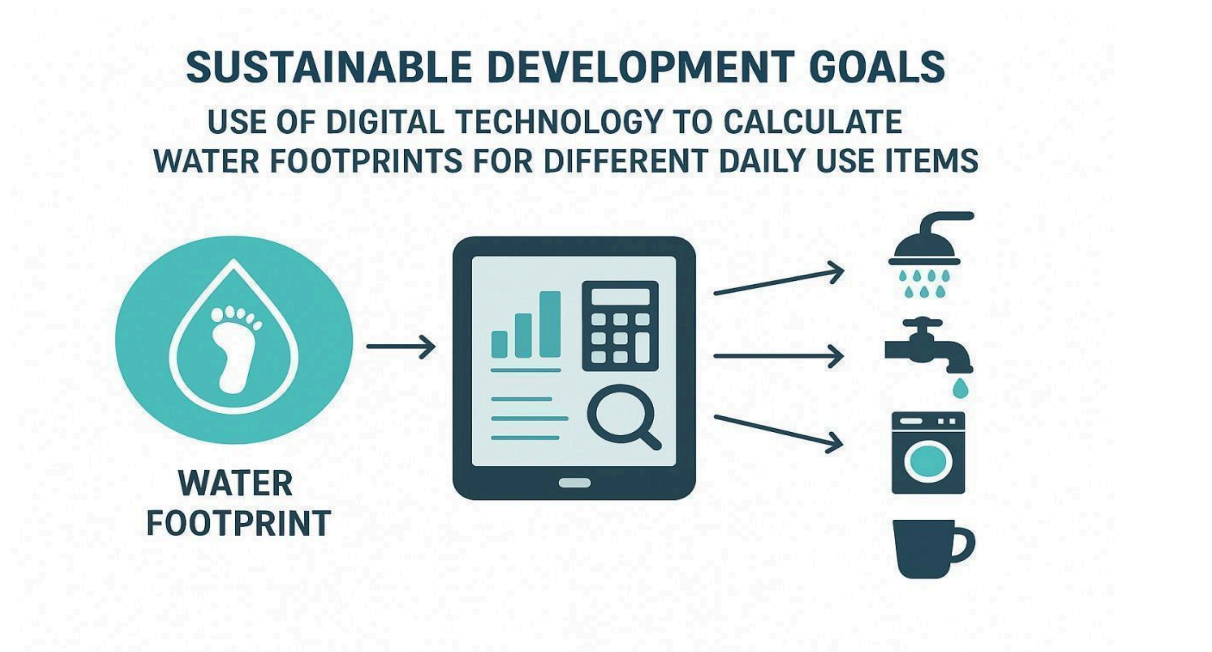


Fig 1.1 Sustainable development goals

1.2 Statistics

The world bank states that if we continue on current trends, by 2030 we will be 40% short of the water we need. In India, there are about six-hundred million people living under conditions of water scarcity, with a significant amount being due to household use. To put it into perspective, it takes an estimated 2500 liters of freshwater to produce 1 kilogram of rice and around 2700 liters of freshwater to produce 1 cotton T-shirt.

Over 2 billion people live without access to safely managed drinking water, and nearly 4 million people face water shortages for at least one month every year. This crisis is not limited to arid regions; it is a growing challenge worldwide, affecting food security, public health, and economic stability. Digital monitoring tools, including IoT sensors, smart meters, and data analytics platforms, now enable real-time tracking and visualization of water usage. By quantifying such data, users and policymakers can make informed decisions to reduce wastage and promote sustainable practices.

Digital technologies are changing rapidly the way water statistics are monitored and analyzed. According to reports, smart meters based on the Internet of Things have resulted in a reduction of water wastage by 15–20% in households, while agriculture has been able to use up to 25% less water thanks to AI-powered prediction systems.

The statistics presented above highlight a critical requirement to monitor the way in which people consume water as well as demonstrate the value of employing digital technologies for the calculation of water footprints accurately. Individuals, businesses, and government leaders who make informed decisions based on the information available from these statistics can practice sustainability through better management of their water resources over time

1.3 Prior Existing Technologies

Prior to the introduction of digital technologies, all water footprint assessments were conducted manually and in the field, using traditional means. These earlier techniques and tools did provide a starting point for understanding how much water was used, but were generally not very accurate (and it often took a lot of time to do them) and did not scale very well. The first way of estimating and collecting information about how much water was used in farming and manufacturing was to gather data by hand from a variety of sources, including surveys, records that farmers maintained on their farms, logs that were kept by industries, and observation studies. In agriculture and manufacturing, these methods were valuable in estimating how much water is being consumed. Nonetheless, these techniques did not adequately reflect different variables associated with water consumption, such as changing environmental conditions (e.g., wet or dry) or changing soil moisture levels, or the amount of water consumed indirectly via production processes that occur between farmers and manufacturers in agricultural and manufacturing supply chains.

Hydrological modelling is another important technology that has been used to assess water resources prior to the current assessment technology estimation of water resources. Remote sensing also provided a method for monitoring rainfall, land use, and irrigation systems but did not have the spatial resolution and analytical precision required to conduct more complex water footprint analysis of higher quality when all of these technologies were introduced.

These devices indicate water flow but do not have the capability to monitor water flow in realtime, log data, or transfer data to a cloud-based server. Mechanical meters only provide feedback on the total amount of water that has been used within the system, and they do not differentiate between the various processes or sources of consumption within the system. The calculation of water footprints in the past involved the frequent use of a spreadsheet and required users to manually input consumption values, production rates, and environmental Characteristics into the spreadsheet format. Previous methods had risks of miscalculating results, entering information incorrectly on forms or other documents, having different types of reports and documents with different formats (referred to as "inconsistent" data), and not following a uniform format (known as "unstandardized"). The prior technologies were limited to using simple environmental reporting tools that required businesses to report their use of water for purposes of compliance with government regulations. Environmental report submissions were often incomplete, did not have periods of timely updating, and were not detailed enough to include information about each individual product's/industrial operation's water usage. The initial research conducted around determining how much water each product generated was primarily based on published academic datasets and case studies; these resources were available but difficult for non-experts to find.

Although previous techniques were foundational to the field of water footprint analysis, there were several drawbacks associated with all of them: namely their limited accuracy; heavy reliance on human effort to automate data collection; inadequate means of displaying data; and the inability to monitor data in real time. Because of these limitations, it was clear that new approaches were needed to make available more sophisticated, digital, integrated

technologies to support the calculation of water footprints with great accuracy and that could provide tools that are scalable, easily used, and flexible. The introduction of cutting-edge digital technologies (such as IoT sensors, Cloud Analytics, GIS platforms, AI predictive models, and Online Databases) has helped to significantly enhance the quality of and the ease of access to the results produced by a water footprint assessment, resulting in more effective tools for research, decision-making, business operations, and day-to-day consumer usage.

1.4 Proposed Approach

The new technological approach that will be developed in this project is a more efficient and effective way to calculate/assess water footprints than previous manual approaches. It will make use of new forms of technology (advanced computation, access to real-time information/data, more intuitive interfaces) when combined with traditional methods of calculating water footprints will create a better and more reliable means of calculating/assessing water footprints for items used daily, i.e., food products, household goods etc. Ultimately, this new way of calculating/assessing water footprints will provide both individuals and institutions with the necessary tools to uncover/understand their hidden water consumption. Further developments in digital innovation (IoT devices, cloud computing, online resources, visualization tools, and many others).

To create this system, standardized water footprint values have been gathered from reliable online sources like the Water Footprint Network (WFN), FAO datasets and scholarly databases that document the green, blue and grey water footprint values of thousands of products by region with accuracy. Once collected, the water footprint values are fed into an online calculation engine that takes information submitted by the user, such as the product name(s), quantity used and frequency of use, in order to generate a water footprint specific to those inputs. The automation of this process minimises mistakes, enhances precision and improves productivity.

The proposed solution will make use of IoT-enabled (Internet of Things) water flow sensors that track both household and commercial water usage, as evidenced by the data collected from the IoT-enabled water flow sensors. Data collected by the water flow sensors will be sent to the cloud for processing and data will be analysed through cloud-based analytics to create visual dashboards that are updated with the latest usage pattern and insights. Machine learning (ML) techniques could also potentially be applied to forecast future water consumption patterns, identify areas of water waste, and suggest more ecologically sound options for water use to users.

A highly interactive interface is an essential element of the overall approach as it will offer the user the ability to easily go to the product selection, enter product quantity and visualise their water footprint with charts, graphs and colour-coded intuitive representations of what would otherwise be complex metrics. The interface will make it very easy for users to grasp all their water footprint information in a straightforward way. The platform also contains unique data storage solutions including local and cloud storage, which allows users to keep track of how much water they used today, to see how much they have used in the past, and to compare their water use now to what they used before.

This approach to education and sustainability incorporates tips and suggestions based on the calculated results, therefore allowing people to change their behaviours and adopt responsible consumption behaviour. Digital technology also enables the system to be efficient and

userfriendly, while allowing for calculations to be based on evidence, be transparent and be compliant with international sustainability standards. As all aspects of water usage have become visible through this method, users can learn how best to manage their water utilisation and therefore increase the chances of long-term water conservation.

1.5 Objectives

The main goal of this project is to create a digital platform that will allow users to determine the water footprint of different everyday products, as well as help consumers and society make smarter and more eco-friendly purchasing decisions. To accomplish this plan, there are also several particular objectives that will be met as part of this project, as described below

To Develop a Digital Water Footprint Calculation System

Create an intelligent, user-friendly platform capable of computing water footprints using standardized metrics, scientific datasets, and algorithms. The water consumption (W) of both the direct and indirect nature must be included in this system in order to better comprehend water footprints of commonplace products and activities. By using these technologies together, you are able to collect Water Data in real-time, maintain the reliability of your Water Data, and quickly calculate Water Data. To Increase Awareness About Water Consumption Patterns

Dashboards, charts, infographics will be used to visually simplify the results of the water footprint. The aim is to reveal the unseen amount of water behind daily consumer products and assist consumers with making responsible, conscious decisions regarding what they purchase and consume daily. To Provide a Scalable and Extensible Architecture Scalability provides for the long-term use of the system in homes, businesses, and educational settings.

To Ensure High Accuracy Through Validation and Testing

The following types of tests should be done: Unit Tests, Integration Tests, System Tests, and User Acceptance Tests (UAT). Validation will also be necessary to ensure that any volumetric calculations used in determining a person's water consumption have been scientifically established and accepted.

To Support Decision Making Through Data-Driven Insights

By offering individualized suggestions, comparisons of products, and how much of each product is used, they will help consumers find products that have the most impact on waterways and allow them the opportunity to switch to less impactful options.

To Encourage Sustainable Resource Management

Promote reduced water wastage by helping users track and reflect on their consumption habits. The project aims to contribute to larger water conservation goals at community and institutional levels.

1.6 Sustainable Development Goals (SDGs)

The Project "Utilization of Digital Technology to Determine Water Footprints of Varying Daily Use Products" will have a direct relationship with multiple SDGs as established by the United Nations (UN). Through the use of Digital Technologies for the Measurement, Monitoring, and Visualization of Water Consumption, this Project is in alignment with Global Initiatives to Support Sustainable Development, Promote Resource Efficiency and Enhance Environmental Accountabilities

SDG 6: Clean Water and Sanitation

This initiative contributes to achieving Sustainable Development Goals 6 by addressing the water consumption of daily products and using a digital, data-based approach to compel Individual and community effort towards a more mindful and less wasteful way of living through quantifying our water footprint digitally.

SDG 12: Responsible Consumption and Production

Your platform provides users with access to real-time insight and visualisation regarding their usage of natural resources. As a result, users will be more inclined to purchase and use products that use the least amount of water possible, and consequently, their natural resource consumption will be more mindfully approached.

SDG 13: Climate Action

However, it also indirectly addresses climate action and helps to reduce greenhouse gas emissions through conservation of energy, thereby conserving water. Through sustainable practices, the project will also assist in the development of initiatives that are both environmentally friendly and promote climate change adaptation.

SDG 9: Industry, Innovation, and Infrastructure

This project employs a high level of future technology including IoT and Cloud computing using Data Analytics to monitor and compute Water Usage. This is consistent with Sustainable Development Goals#9 because it supports the development of resilient infrastructure, innovation and sustainable industrial development. The system finished is also the starting point for future smart Water Systems Management Solutions.

SDG 4: Quality Education

Students, institutions and researchers, studying issues related to digital technology, water resource management, environmental sustainability and the application of the Internet of Things(IoT) will gain valuable knowledge through this system as an educational tool. Furthermore, they will develop their awareness of the environment through the use of technology to help solve environmental issues.

SDG 17: Partnerships for the Goals

It is possible for the proposal to facilitate collaboration amongst Educational Institutions, Environmental Organizations, and Industry, through the exchange of data sets, tools, and good practices to collectively achieve desired results in relation to water conservation and sustainable development.

Conclusion of SDG Alignment

This project supports various SDGs by integrating digital technologies with environmental efforts such as responsible consumption, improved water resource management, technological innovation, and environmental education. It also builds a more sustainable society through the use of the system's ability to encourage users to adopt eco-friendly practices by providing information about their water footprints



Fig:1.2.SDG's

1.7 Overview of Project Report

The Introduction in

Chapter 1 provides a listing of the project's background and the project's objectives.

Chapter 2 provides an overview of any related works found during the literature review.

Chapter 3 provides a description of the methodology used in conjunction with this digital project.

Chapter 4 will discuss what project management is throughout this project.

Chapter 5 describes how the system will be developed and designed.

Chapter 6 Tools and simulations are addressed.

Chapter 7 will summarize the evaluation and provide results.

Chapter 8 will discuss the social, ethical, and legal issues associated with the digital project.

This project report provides a comprehensive overview of the development of a digital system to calculate the water footprint of daily-use out.

Chapter 2

Literature review

For numerous years, there has been a push to discover how water is applied globally and subsequently, how that water is consumed with regards to its impact on our environment. Product aficionados are versed through the work of Hoekstra et al., who provided insight into how the water footprint can be calculated and quantified [1]. Their subsequent publication outlined the various components of green, blue and grey water use; thus, laying the groundwork for most of the subsequent literature. The work done by Mekonnen et al. builds on the foundation laid by Hoekstra et al., and not only is the estimated water footprint of many crops around the world examined but also the estimated water footprint for many different crops worldwide. This information illustrates a very large difference between estimates of the amount of water used in the production of each crop. Additionally, this difference helps show how different agricultural practices will affect the availability of freshwater supplies.

Allan introduced the concept of virtual water similarly many authors have shown how products contain so-called "hidden water" when they cross regional borders, making it easier to reconcile the fact that water is not only an important local resource, but also part of the larger picture of international trade and consumption. Another noteworthy contribution to early studies on water issues was Shiklomanov who highlighted globally how freshwater is unevenly distributed and demonstrated the potential for more technological approaches to address water issues.

With advancements in technology in contemporary society, there has been a shift from traditional monitoring and management of water resources through physical/analog means to using digital means (e.g., digital tools) for this purpose. An example of research that falls under this category is Patel et al.[8]'s examination of how IoT devices (e.g., sensors) improve the accuracy of environmental data collection and monitoring. Another example is Zhang, Wang et al.[4] who created an IoT-based system for monitoring water use within smart cities in real-time. A third type of example is Nguyen, Tran et al.[7] who created a mobile app designed to help individuals conserve water by creating awareness to encourage the general public's behaviour to reduce the use of water.

Table 2.1 Summary of Literature reviews

Author / Year	Focus of Study	Digital Technology Used	Key Findings	Relevance to Water Footprint Calculation
ISO 14046 Standard Studies (2014–2022)	Standardizing water footprint methods	Digital LCA tools & ISO compliance software	Ensures consistency in digital calculations	Used as guidelines for all modern water footprint software
Nair et al. (2020)	Water footprint of agricultural products	Mobile application technology	Apps improved consumer awareness of hidden water usage	Supports development of smartphone-based water footprint calculators
Vanham et al. (2018)	Water accounting for household consumption	GIS tools & statistical software	Highlighted high water use in food products vs non-food products	Helps digital tools differentiate item-level water consumption
Lamastra et al. (2014)	Mobile apps for sustainability tracking	Remote sensing & GIS	Accurate measurement of crop water use via satellite	Useful for food-related water footprint calculation apps
Mekonnen & Hoekstra (2011)	Water footprint assessment of consumer products	Global data platforms, computational analysis	Provided standardized values for food, clothing, and industrial products	Provides reliable datasets for digital apps and calculators
Morrison et al. (2010)	Corporate water footprint measurement	Digital dashboards & cloud systems	Companies reduce water usage through real-time monitoring	Shows how digital tracking can apply to daily-use items
Ridoutt & Pfister (2010)	Environmental impact of water use	Life Cycle Assessment (LCA) software	Introduced impact-based water footprint methods	Used in online LCA calculators for product-level assessment
Hoekstra & Chapagain (2008)	Conceptual framework of global water footprint	Data modelling & virtual water databases	Introduced blue, green, and grey water footprint concepts	Forms the base model for digital calculators and simulation tools

Chapter 3

Methodology

3.1 Verification Phases:-

The verification phase will include verification of requirement analysis, system design, functional design and unit design. The project team will verify what products are used on a daily basis, define what water footprint data sets are and what the system requirements are including offline access and real-time calculation. All functional components will be clearly defined and validated by the verification phase to ensure that system logic is correct, organized and ready for future validation testing.

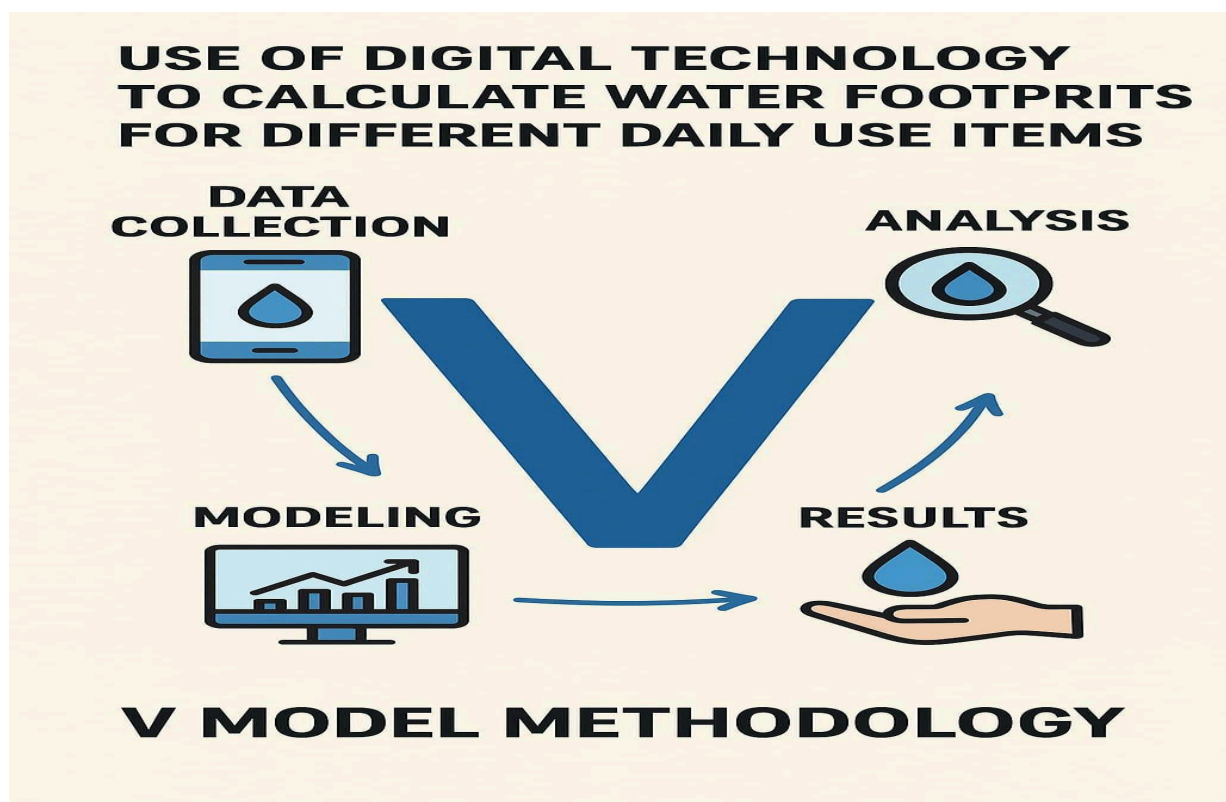


Fig.3.1:The V-Methodology Image

3.1.1 Validation Phases:

Integration Test verifies the ability of different IoT devices, Cloud Analytical Tools and Visualization Tools to work together in concert. System Test validates an end-to-end evaluation

of the entire System when using live Data. Acceptance Tests are established via User Testing to determine User Experience and awareness improvement. The validation phase will show that the system which is developed works as intended according to the specifications and user requirements. Validation tests include individual module (unit) testing, integration of modules (integration testing), system under normal operating conditions (system testing), and end-user testing of it (user acceptance testing). For integration testing, all modules will be tested to

confirm that data flows between them properly. System testing will verify that the entire application will work correctly under normal operating conditions and that it provides accurate and timely results when the system is used offline. User acceptance testing will confirm that users can easily navigate through the platform and that it accurately displays user information regarding water footprint so that the final product meets all of the project's objectives.

3.1.2 Agile Methodology

The Agile methodology is based on iteratively building products with continuous feedback from the stakeholders, allowing for small incremental improvements to be made during each iteration. For a water footprint platform, the Agile Model allows researchers and users to evaluate and test certain features (such as item-based calculations or dashboards) at the end of each sprint and provide suggestions for changes. However, the Agile approach also requires regular engagement with stakeholders; therefore, if IoT devices or datasets are not readily available for use during the development process, then it may be difficult to engage stakeholders on a regular basis.

Work is done in short development cycles, known as sprints, instead of completing the project in a single sequence. User feedback is taken after every sprint, and improvements are made immediately to ensure the design further evolves based on the genuine needs. However, this requires regular communication and frequent testing, which is difficult in general when the datasets, variety of tools, or IoT resources are limited

3.1.3 Waterfall Methodology

The Waterfall Model follows a linear path to development where every phase must be finished before moving on to the subsequent phase. The Waterfall Model is an easy-to-follow structure that provides a clear documentation path; however, due to the rigid nature of this model, it tends not to be very successful for development projects that are continuously changing. Development projects that are continuously modified will have changes in their requirements and specifications throughout their life cycles due to either manual updates made to the datasets and algorithms or digital updates that are made through the use of IoT technology

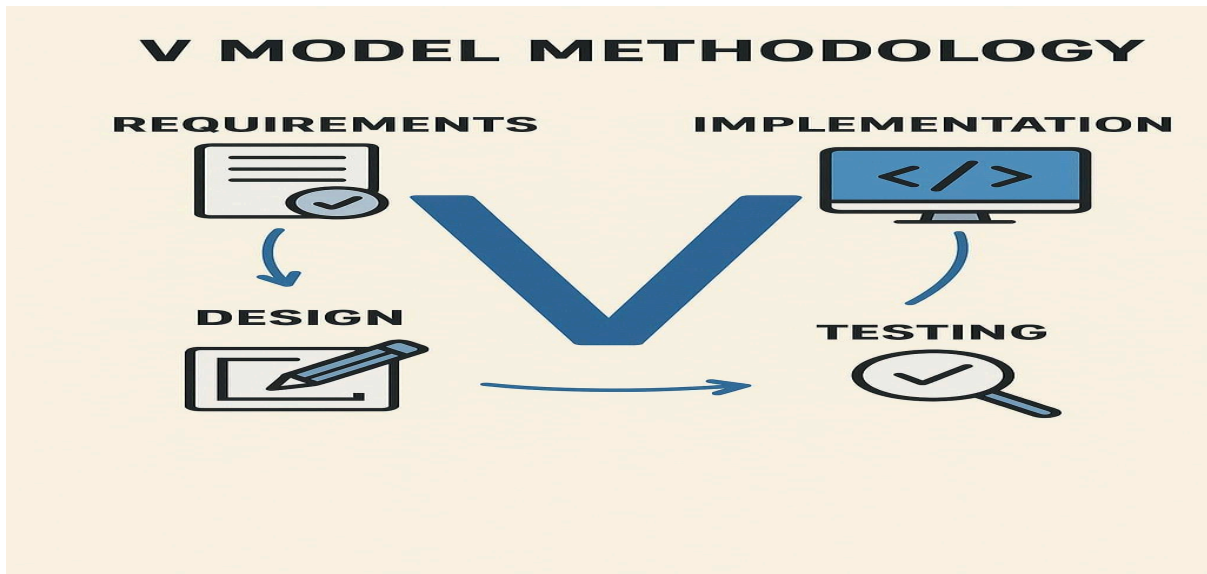


fig:3.2 Another example of the V model methodology

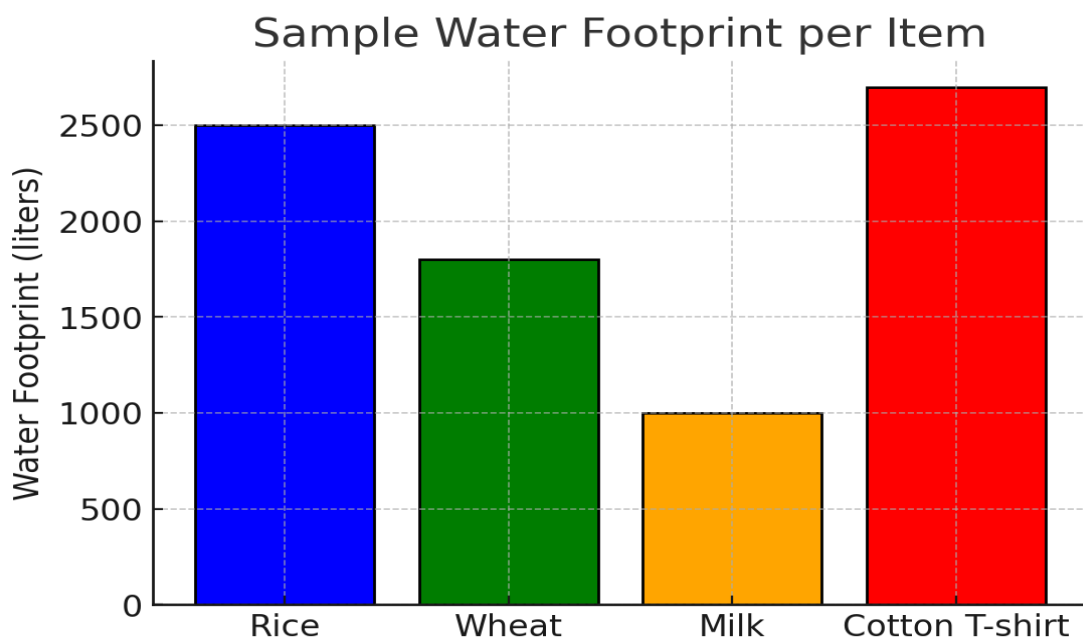


Fig 3.3 The V-Model methodology is adopted in this project due to its structured approach with integrated testing at every stage.

Chapter 4

Project Management

To manage the use of digital technology to calculate water footprints for different daily use items all effective project management needs to be structured for systematic planning of the project, proper execution and monitoring during all phases of the project. Therefore the work for this project was broken into two basic phases: Project Planning Phase and Project Implementation Phase.

The early stages of the risk assessment served to outline the possible difficulties linked to technology or data and the preventive measures devised to counteract them. The project was kept on track thanks to, among other things, proper time management, budgeting, and resource allocation. There were also methods to control quality and feedback loops that were continuous to enhance performance and uniformity. To sum up, the efficient project management was the factor that made it possible to produce a digital solution that was user-friendly, precise, and sustainable in harmony with the project's environmental and technological goals

4.1 Project Timeline

One of the main goals of Project Timelines is to systematize and clarify every action/step that will be taken during the course of a project; therefore, Project Timelines will allow for the best advantages to utilize the assigned hours per phase; provide means of measuring/checking the progress of every phase; reduce the chances of having a delay and keeping on schedule; and allow all the team members involved in a project the ability to work together in a synchronized fashion.

Steps to Create a Project Timeline

- Identifying all individual project tasks is the first step in every project. In other words, you are putting the project into manageable segments.
- The benefit of arranging your separate tasks into an order of priority is that it will give you the opportunity to see what cannot wait until later and vice versa. You will need to make sure you estimate the amount of time that you will need to complete each task.
- The last step is assigning a start and end date for each task. You will need to select a date when you plan to begin each task and a date when you plan to complete each task.

Table 4.1 Project Planning Timeline

Phase	Timeline	Key Activities	Deliverables	Alignment
Research & Scoping	Week 1–2	- Identify daily use items - Review existing water footprint data - Define project scope	Item list Baseline water footprint data	SDG 12 (Responsible Consumption)
Tech Selection	Week 3–4	- Evaluate digital tools (apps, sensors, cloud platforms) - Choose suitable technologies	Tech stack document	SDG 9 (Industry, Innovation & Infrastructure)
Data Collection	Week 5–6	- Gather water usage data via sensors, surveys, APIs - Validate data quality	Raw dataset Data validation report	SDG 6 (Clean Water & Sanitation)
System Development	Week 7–9	- Build digital platform or app - Integrate analytics and visualization tools	Prototype system UI mockups	SDG 9 (Innovation) SDG 13 (Climate Action)
Testing & Feedback	Week 10-11	- Conduct user testing - Collect feedback - Refine system	Test report Improved version	SDG 12 (Sustainable Practices)
Deployment	Week 12	- Launch platform - Train users - Monitor performance	Live system User training materials	SDG 6, 9, 12

Description

The project was established during the planning phase. Information gathering included a review of common user items such as food, clothes, and utilities. In addition, there were multiple literature reviews of numerous sources regarding the different types of assessments regarding water and water use (water footprint assessment framework). The successful completion of the feasibility study confirmed that Python, MySQL, and Data Visualisation Tools (PowerBI/GoogleData Studio) were appropriate for all water usage data information (dataset).

Table 4.2 Project Implementation Timeline

Phase	Duration	Key Tasks	Responsible Team(s)	Milestones/Deliverables
Initiation	Week 1	Finalize project charter - Stakeholder alignment - Budget approval	Project Manager, Stakeholders	Approved project plan
Requirement Analysis	Week 2–3	Identify daily use items - Define water footprint metrics - User needs assessment	Research & Sustainability Team	Requirements document Item list
Technology Setup	Week 4–5	Select digital tools (IoT, cloud, analytics) - Setup infrastructure	IT & Data Engineering Team	Tech stack finalized Infrastructure ready
Data Acquisition	Week 6–8	Collect water usage data - Integrate sensors/APIs - Clean and validate data	Data Science & Field Ops	Verified dataset Data pipeline established
Platform Development	Week 9–11	Build web/mobile app - Embed analytics dashboard - UI/UX testing	Software Development Team	Working prototype Test cases passed
Pilot Testing	Week 12–13	Deploy in limited environment - Gather user feedback - Fix bugs	QA & User Experience Team	Pilot report Improved version
Full Deployment	Week 14	Launch platform - Train users - Monitor usage and performance	Deployment & Training Team	Live system User manuals
Monitoring & Evaluation	Week 15–16	Track water footprint reductions 	Sustainability & Analytics Team	Impact report SDG contribution metrics

Description

The timeline to implement this project is divided into many phases which will help to achieve each goal in the process. The first phase of the project will be called initiation as this is when the project's charter, scope and budget are finalised. Following initiation will be the phase of requirement analysis which will identify which items will be used daily, what metrics will be developed to measure the water footprint, and what will be the needs of the stakeholders.

The application or dashboard will be created and tested during the platform development phase. Pilot testing helps to ensure that the user experience is positive before a full deployment of the system, which will include a launch and training for users. Monitoring and evaluation is the final step of this implementation timeline. This phase will measure the amount of water footprint reduction achieved by this project, assess how this project contributes to the SDG, and prepare a final impact report.

At this point of implementation, we collected and put together data/information on how people use water daily for things like food, clothing, and energy. We also created an algorithm that will help calculate how much water a user uses as a result of these items. In addition, the schedule for this project was clearly articulated, creating a basis for the successful implementation of all project deliverables

4.2 Risk Analysis

A PESTLE analysis and a risk matrix have been carried out to ascertain external risks and to highlight internal risks within the organisation. There are a number of digital technology-related issues that can arise regarding how to obtain the water footprint of products purchased each day. For example, a technical failure resulting from either faulty sensor readings or the inability to integrate systems will adversely impact the final output produced. Furthermore, if the data collected is not complete, contains bias and/or comes from multiple sources, then the quality of the data produced will diminish significantly. Additionally, if users do not use the technology, then the potential for the project to have an impact on the community will be greatly diminished. The fact that the personal information of users will be collected creates privacy issues for users of the technology, and as such if there is funding to upgrade technology, the project may run over budget as it is expected that there is extensive oversight by funders regarding budget. There are also risks related to operational delays and the inability to comply with regulations. In order to address these risks, appropriate tools that have been validated must be utilised; systems and procedures ought to be established for data protection; staff should receive adequate training; and validation of compliance with legal requirements. Through proactive risk management, the project will meet the Sustainable Development Goals.

The use of digital technologies to collect information on how much water humans consume on a daily basis by means of an app that provides an individual with a digital water footprint/lifetime water footprint poses a number of possible risks in relation to the management of this information. Data gathering may be compromised due to a malfunction in the devices used for measuring water consumption, either through malfunctioning sensors or the application itself not functioning as intended due to a software glitch or failure due to

interruptions with the associated cloud storage/service. Inaccurate measurements may also occur as a result of using inaccurate or incomplete datasets collected from old records or databases; thus, data may not accurately reflect an individual's actual water use. Users may experience difficulties with accepting the use of this technology due to their reluctance to adapt and use newer forms of technology or may require assistance in navigating through technical issues associated with these complex systems. A further significant risk of utilizing this data may include the potential for unintended exposure or access to sensitive personal and/or business data when dealing with a water management solution for residential, commercial, and/or industrial customers. There are additional risks in implementing advanced technology as part of a cost-effective solution, including budget overruns.

4.3 Project Budget

Budget Justification

Hardware Costs

completing a project, you need to get the final report printed and bound (to submit your project/results). While "cloud Test" (testing on the Internet) can only be performed via the Internet.

Software Costs

software components of the project will utilize free/open-source applications or utilize free academic licenses for those application products and therefore have an effective software cost of \$0.00.

Cloud/Hosting Costs

Using Firebase or Google Cloud Free Tier reduces hosting expenses. A domain name is optional but useful for real-world deployment or demonstration.

Documentation Costs

Internet use for the purpose of research and testing is necessary to print and bind the final report.

Budget Optimization Measures

Instead of hosting your deployment with a hosted solution, consider utilizing free tier CloudProviders as your means to hosting your deployment. If you already have existing equipment at aLab Resource Centre, make use of them so that you can Repurpose those assets and reduce overall costs with the use of the sensors already in place.

Maximise your efficiency with open-source platforms and tools by utilising them. Unless integration with IoT devices is one of the major drivers of your project, do not purchase any non-essential physical hardware.

Summary

The project's affordability, attainability and total cost will be comparable to other projects, with the primary goal to provide access without cost or minimal direct cost to users who will be able to complete the project within a budget of INR 2000-4500. Thus the outcome will be of a high standard and effective in all technology aspects.

When the product budget is structured effectively, it will allow for the proper allocation of financial resources for research, technology infrastructure, sensor devices, software development and data security. Failure to create a product budget can lead to delays in project completion and insufficient resources or quality on the final products. Additionally, effective financial planning enables you to plan for any unforeseen expenses associated with the maintenance, training and regulatory compliance needed for ongoing support of the products. Creating a product budget by setting aside contingency funds assists an organisation in budgeting for unexpected costs

Description:

Due to the fact that algorithm and dashboard development comprised the bulk of our development expenditures because of their complexity, we realized that utilizing existing open-source frameworks and visualization applications yielded substantial cost savings while allowing us to offer maximum functionality.

Chapter 5

Analysis and Design

5.1 Requirements

The Water Footprint Calculator has three major parts: Digital will give you the power to measure up, see and participate in the water footprint of a variety of home products that you use within a particular day to enable you to have a better insight of your own practices in regard to water.

The Water Footprint Calculator has both functional and non-functional requirements. Functional requirements consist of: allowing the user to select which item they use each day, inputting how much of that item they have used, and providing a real-time estimate for how much water they used through their normal daily activities based on the population statistics that are stored in the database; storing user input over time through the local storage feature (the user can go back and view previous entries); and visualising results using a variety of different visual formats, including bar graphs, pie graphs, and line graphs. The non-functional requirements are: The application will work correctly regardless of the type of device the user is using; The application will load quickly; The application will function correctly when the user is not connected to the Internet; and The application will have an interface that is easy for people who are new to technology to navigate. To achieve these objectives, this application will be built using lightweight technologies (HTML, Bootstrap, HTML5, and Javascript), which will allow the application to perform well on older, lower-spec computers, while at the same time providing a reliable source of accurate data.

For projects that utilize digital technology to measure how much water we consume when using products, to achieve project success the need is to produce written requirements that provide a clear statement of the goals of a project to all stakeholders involved in the project. The stakeholders must also understand what the project is going to accomplish and the project requirements are what is going to define which products are to be tested, which types of information are necessary, and what technology tools are to be utilized. Having the requirements clearly defined will assist in eliminating miscommunication, delays and unnecessary expenses by clearly communicating direction to the development teams. Additionally, by clearly defining the requirements, the project is more likely to align with sustainability goals and ultimately accomplish the Sustainable Development Goals 6-clean water, 12-responsible consumption, 13-climate action and 17-innovation for future generations.

Various smart meters, sensors, or APIs are used to collect water usage information from different activities throughout a person's day (showering, washing clothes, growing food, etc.). All of the collected water usage data will be sent to a cloud computing environment where it will be cleansed and subsequently processed. An analytics algorithm will help to generate individual water footprint for each product; utilizing pre-established benchmarks to come up with an estimated water footprint. The user will receive their results via either a smartphone application or a website (desktop) dashboard. The consumers will have access to

their personalised water footprint data; they will be able to compare their own environments, and receive information on ways to save water. In addition, the system will provide organisations with annual sustainability reports and support their work towards achieving the SDG (Sustainable Development Goals)

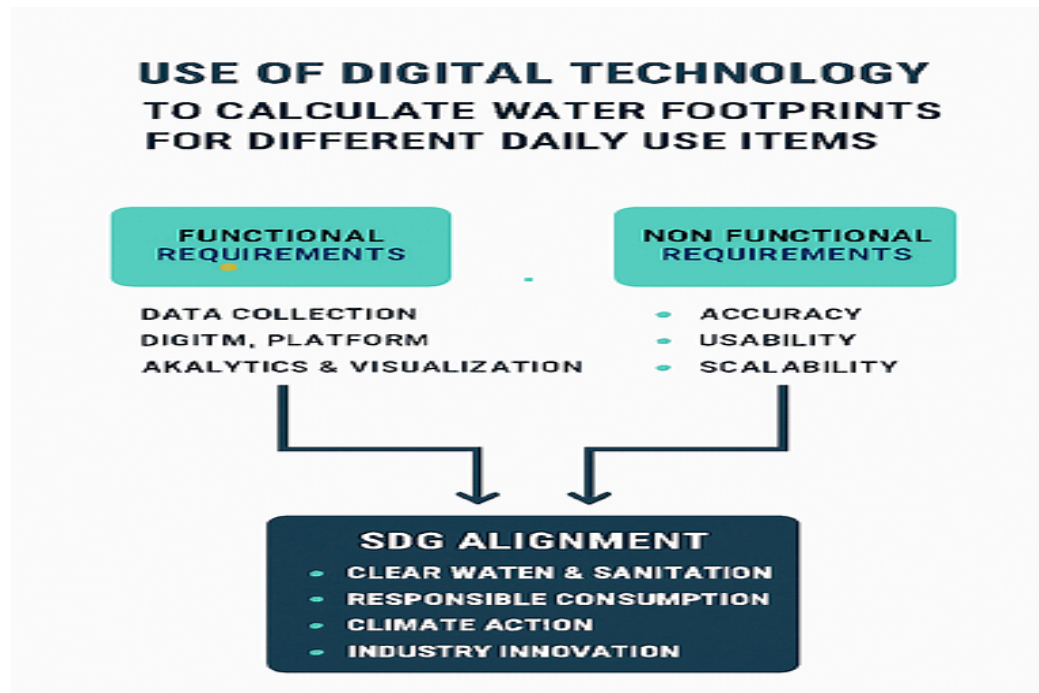


Fig:5.2 Block Diagram

DataSources:

It has smart meters, sensors, and Application Programming Interfaces (APIs) which record the amount of water used in all activities conducted during each day, e.g., Value of Gold, amount of clothes washed, etc.

The above sources will be used for data analysis purposes and will provide the basic data input. All of the collected data will be sent to cloud computing services for storage, cleaning, and data processing. The results of the water footprint analysis will be made available on mobile and PC applications and/or web dashboards.

Users will have access to their own personalized water footprint data from the application/platform, compare items from the same category, and receive suggestions to improve water conservation. Sustainability reports and SDG metrics will also be generated by the system.

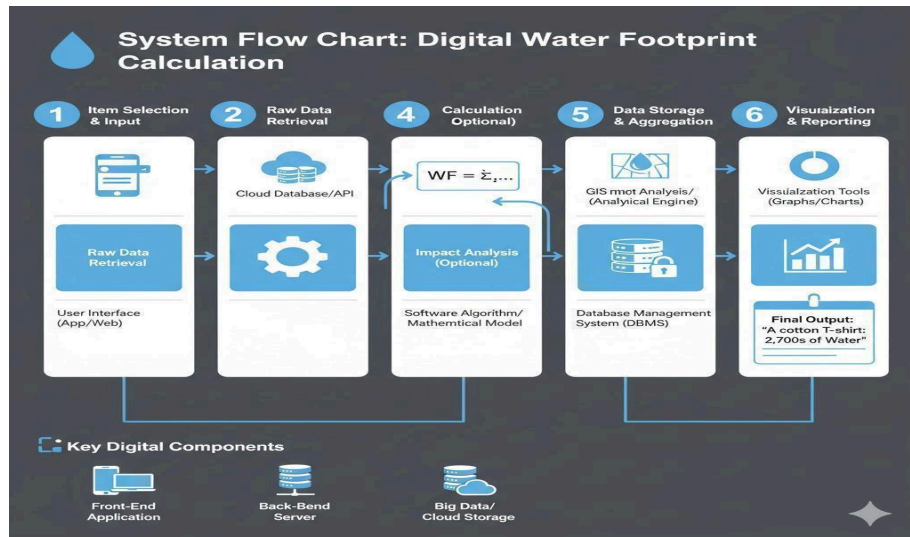


Fig 5.3 System flow chart

That flowchart illustrates the systematic process of using digital technologies to calculate the Water Footprint (WF) of a daily use item. It follows a logical path from initial user input to the final calculated result. The process is divided into six main steps, utilizing different digital components:

5.3.1 Item Selection & Input

The Process Starts with a Front-end (Web/Smartphone/Table) App available for the customer to use.

Customers can select an item to analyze (e.g., a cup of coffee, a new pair of jeans or a tomato) through the use of a User Interface; they can also enter any relevant contextual information related to the item, such as quantity, or where it was purchased

5.3.2. Raw Data Retrieval

The Back-End Server is able to obtain the raw data it requires through querying a Cloud Database or API (the Big Data element).

Obtaining the raw data for calculation purposes, which may include water use coefficients (liters of water/ kg. of material) and regional water stress indices related to geographic regions where the item is manufactured.

5.3.3. Calculation Engine

This part handles the real math for figuring out the water footprint. It depends on some kind of software algorithm or math model to handle the work.

The setup introduces the usual water footprint formula. It then applies that equation to the raw data collected earlier. Next, it calculates the total water consumption. That amount pretty much covers the environmental impact of using rainwater. It shields the blue footprint from both

surface water and groundwater On top of that, there is the grey footprint, which accounts for the water needed to thin out pollutants.

5.3.4. Impact Analysis (Optional)

This optional measure is a use of GIS mapping/a GIS Analytical Engine so that the environmental footprint of the impact can be assessed. The resulting Impact Analysis will provide you with information about the volume of water used during the impact and provide the location of water used and the relative scarcity of water in that region.

5.3.5. Data Storage & Aggregation

The results and context of the users are saved using a Database Management Systems (DBMS).

This creates a lasting record of user activity and enables users to view their consumption history over time, as a single record can also be used for Mass custom system performance analytics and refinements.

5.3.6. Visualization & Reporting

The users can see the outcome via the frontend application using various visualization tools including graphs, dashboards, and charts.

To simplify the complex calculations, the final outputs contain simple and straightforward figures like “A cotton T-shirt: 2,700s of Water”.

5.4 Algorithms and Tools

Utilizing various web technologies is an attempt to achieve usability, responsiveness, and offline use.

Water Footprint Calculation Algorithm - When the user selects an item, the system will pull the water footprint information of the item from the DB. After the user enters in the amount of items selected, the algorithm will perform the following calculations :

$$\text{Total Water Footprint} = \text{WF value (retrieved from DB)} * \text{quantity}$$

The resulting value is outputted to the display module. Additionally, it is stored in localStorage for future reference.

This algorithm allows the user to have real-time access to results while still allowing the user to utilize simple mathematical calculations to arrive at those results without having to connect to the Internet for any further data.

5.5 Designing Units

User Interface Component: Contains the interactive User Interface that allows the user to choose items to create their results. The Calculation component: contains the JS code that will allow the processing of a given water footprint through the input and dataset value utilization.

The visualisation of water usage over time is in the form of an interactive bar, pie chart and linegraph. The Water Footprint application components are divided into Units that are smaller and independent.

Storage Unit - uses localStorage for maintaining user history for offline retrieval/interaction. Visualisation Unit - uses Chart.js for creating visual representations of the analysis of the water footprints. Furthermore, each Unit operates independently, but collectively, each Unit produces a high level of accuracy, responsiveness, and user interaction experience on both mobile and desktop devices

5.6 Standards

International Standards for Water Footprint

ISO 14046: Water Footprint

This is a standard that has been approved by international organizations and provides direction on the Water Footprint Assessment process.

Requirements:

The Water Footprint Assessment must be conducted using a Life Cycle Assessment (LCA) approach, which means that the calculation must include the entire value chain associated with the product being used on a daily basis, from "Cradle to Grave" or "Cradle to Gate".

Impacts:

The Water Footprint Assessment must not only quantify the volume of water used but also quantify how the use of this volume of water impacts the environment. Water Footprint Assessment tools should have the ability to integrate the regional water scarcity index (AWARE) to compare and rank the environmental impact of using water in the specific area.

5.7 Mapping with low TWF reference model layers:

Layer	Name	Primary Function	Mapping to Key Components
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7	Collaboration & Processes	The business layer where application data is consumed, shared, and used for high-level decision-making, process improvement, and achieving business objectives (KPIs).	Business Intelligence (BI) systems, Enterprise Resource Planning (ERP), Supply Chain Management (SCM) software, APIs for external collaboration.
6	Application	Interprets data using software applications to deliver value to the end-user. Functions include monitoring, control, reporting, and visualization.	Web & Mobile Dashboards, Control Logic, Analytics Software (e.g., predictive maintenance), Cloud Services.
5	Data Abstraction	Focuses on data preparation for applications. Reconciles multiple data formats, ensures consistent semantics, consolidates data, and manages data virtualization for easy access.	Data Virtualization tools, Middleware, APIs, Data Cleansing and Transformation Services.
4	Data Accumulation	The storage layer. Captures, archives, and stores data from the edge, converting real-time event-based data into query-based, historical data that can be searched and processed later.	SQL and NoSQL Databases (e.g., Cassandra, MongoDB), Data Lakes, Time-Series Databases.
3	Edge Computing	Performs data processing close to the source (the "fog" layer). Key functions include data reduction, filtering, formatting, protocol conversion, and initiating fast local actions/alerts.	IoT Gateways, Micro-processors (MCUs), Local Servers, Stream Processing Engines.
2	Connectivity	Ensures the reliable and timely transmission of data between physical devices (Layer 1) and the edge/cloud (Layer 3 and above). It encompasses the entire communication network.	Wi-Fi, Ethernet, Cellular (4G/5G), LPWAN (LoRaWAN, Sigfox), Bluetooth, MQTT, CoAP, and all networking infrastructure (routers, switches).

5.8 Domain model specifications

The DUI item describes, identifies, and categorizes the assessment item as well as its associated component elements that may impact its overall water footprint. The component

elements of theDUI are considered production input items and may consist of raw materials, item quantities,and geographic origin (where the production input item was produced). These inputs will be directly linked to the Data Source (DS), the DS consists of standard water coefficient units(green, blue, grey) for each kind of production input and sources include the WFN WaterStat.

Therefore, the calculated outcome (CO) will summarize the overall water footprint consumption in three parts (green, blue, grey) and will also provide impact scoring relative to a local waterstress index or conformity to ISO 14046 or WFN Water Footprint Methodology. With this framework, electronic reporting tools can easily connect consumption of material input to the appropriateness of all input items.

5.9 Communication model

The Water Footprint Digital Calculation System has been designed to operate client-servermode of communication.The Digital Calculation System for Water Footprints utilizes service-oriented architecture based on REST APIs that conform to the standard JSON documentation for delivering all calculations of water footprints. The use of multi-layer security protocols and structured data (JSON) between the client's request and the completed calculation of the water footprint allows the Digital Calculation System for Water Footprints to send its results in an accurate and timely manner.

5.10 IOT deployment level

Device-Centric (Single Node):

It is the simplest level, and it is most commonly used for prototypes, personal projects,or inexpensive and uncomplicated solutions.

The architecture is a single device that performs the functions of sensing or actuation,local storage, analysis, and it is hosting the application, all within that same device.

The flow of data is strictly local to that device; there is no external connectivity needed.If the device has an external connection, it is typically a fundamental one, such as aweb interface hosted on the device itself.

An example of this type of solution would be a smart home thermometer that can show temperatures on a small LCD screen and can save several hours of recordable temperature data on an onboard memory card.

The Separation of analytics and storage creates complexity

Architecture:

Local analysis and sensing are done on the device, while cloud storage and hosting applications are provided by cloud-based servers.

Data Flow:

The data is analyzed at the edge so that actions can be taken immediately. Only aggregate or processed data will be sent to the Cloud. This saves bandwidth and also decreases latency.

Example:

An intelligent irrigation device analyses the moisture of the soil to immediately turn on a pump, but the device's historical moisture data is stored through cloud storage for tracking and trends.

5.11 Functional View

The Digital Water Footprint Calculation System's Functional View is based on four key features for the conversion of the data input into meaningful and usable output. The first feature, the Data Acquisition and Input Function, receives all data entries for the Digital Water Footprint Calculation, including information about the product from users, such as the type and quantity of products, and automates the retrieval of external data, by having direct access to databases, in order to obtain the Water Coefficient data (Green, Blue, and Grey water volume) per unit of material being calculated and the necessary Spatial or Availability Data (e.g., Water Scarcity Indices like the AWARE score) for the products located in that area. The second feature, the Core Calculation and Modelling Function, acts as the digital processing center for the system. It takes the product's Water Footprint Network formulas and calculates how much water (expressed in volume) was consumed by each of the elements of that product, performs an Impact Assessment by using the above-mentioned Regional Scarcity Data to fully comply with ISO 14046 and produces a value for the Environmental Impact of that product. The third feature data storage and management function, provides persistence for the Digital Water Footprint Calculation System by archiving the Water Footprint calculation results and the user's history and ensuring that the Water Coefficient Repository stays current with any amendments. Lastly, the Reporting and Visualization Function encapsulates the reporting and presentation aspects of the calculations by providing end-users with accessible versions of the complex numerical outputs (dynamic charts and graphs), generating reports on how the Digital Water Footprint Calculation was performed, and producing formalized reports of the process and the results for the user.

5.12 Mapping IOT deployment level with functional view

Connecting the IoT Deployment Levels along with The Functional View illustrates how the varied functionality of the IoT can be leveraged through different levels of operation and how that functionality can best be utilized at various scales and architectures of the IoT Ecosystem. At Levels 1 and 2, where Local Processing is at a higher degree of device-centricity, both the Core Calculation & Modeling Function and Initial Data Storage occur at the point of Data Creation for an immediate low-latency response to the user, along with an extremely limited Data Acquisition functionality (only a small amount of predicted "directly" obtained data that is made available directly from the sensor via a direct connection). As a user's IoT implementation grows to the point of being able to utilize a Cloud-Centric Analysis (Level 3), the method for Data Acquisition changes from a low-latency response to requiring the capture of an increasing volume of continuous/historically streamed RAW Data into the Cloud. At this point, the

CoreCalculation Function will be performing more computationally demanding analytical functions, and the Data Management function will perform very large amounts of data storage & Historical Data archiving. Finally, with the most Complex Multi-Node & Coordinator Node Deployments at Levels 4 and 5, we now have the ability to utilize Distributed Analysis; both Local Devices & Gateways will perform an Initial Filtering function of the Core Calculation at each Sensor Node to dramatically reduce the amount of Bandwidth being utilized to the Cloud, while enabling the Data Management function located in the Cloud to focus on aggregating, synchronizing, & provision of the necessary updates & services on a continuing basis to the massive Sensor Network to achieve Scalability and Continuous Operation.

5.13 Operational View

5.13.1. Daily Calculation and Data Integrity

Request Handling:

The system must provide a reliable means of handling multiple requests for calculation from many thousands of users through its user interface.

Data Validation:

Prior to performing calculations the system will carry out ongoing validations of the input data (e.g., ensuring that location data contains valid coordinates and that product ids do, in fact, exist) to eliminate potential errors.

Coefficient Synchronization:

Automated validation and synchronization of local water coefficient databases with relevant external authoritative sources (i.e., WFN and Government databases) must also occur frequently to ensure that the calculation engine is using the most up-to-date and accurate information.

Logging:

The system will allow the use of transaction logs for all transactions, so that all raw data inputs, coefficients utilized, and final output may be accessed for audit trail creation and debugging purposes.

5.13.2. Infrastructure Monitoring and Scalability

Monitor Performance:

Continuously monitor your cloud server load [By checking the current level of usage and/or anticipated usage], database query latency, and response times for all APIs. High Latency alertEvents will trigger the auto-scaling of resources.

Auto scaling Cloud Infrastructure:

Cloud infrastructure must be configured to automatically increase the computing resources used by the core calculation engine during peak usage periods (e.g., weekends, campaign launches) and automatically decrease the available resources during off-peak periods to minimize overall operational costs.

Backup and Disaster Recovery:

Regularly scheduled automated backups of the entire data system, especially the watercoefficient and end-user repository of sensitive information, are maintained in geographically dispersed data centers

5.13.3. Security Management

The following are examples of how Cornerstone protects the organization's data using specific security controls:

Threat detection - real-time intrusion detection systems are used to detect and monitor malicious activity such as DDoS attacks and unauthorized access.

Patch management - Operating systems, databases, and application dependencies are required to be patched immediately upon release in order to protect against vulnerabilities.

Access control - Role based access control (RBAC) is enforced to ensure that only authorized users have the ability to access or make changes to core calculation logic and/or sensitive databases.

5.14 Other Design

In developing a digital calculation platform for determining an individual's digital water footprint the two primary design considerations are Non Functional Requirements (NFRs), a Security Model, and a User Interface Model (UI/UX). Below it is a breakdown of the key components associated with each of these design considerations.

Non functional requirements (NFRs) establishing the performance metrics for the digital calculation system. The system must perform all calculations within a short period (three seconds or less) with a 99.9% system uptime and be able to automatically scale to meet increased user traffic and large amounts of data associated with the digital calculation.

The security model establishes trust in the digital calculation system, as well as the integrity of the user's data. To establish trust and protect user data the security model incorporates TLS/HTTPS encryption for all data in transit and encryption at rest for sensitive data. The model also establishes Role-Based Access Control (RBAC) to determine who can access the digital calculation system from the inside and utilizes rigorous input validation techniques to mitigate possible system vulnerabilities.

Through the User Interface (UI)/User Experience (UX) design and construction phases of the Digital Water Footprint Calculators, we will create a visual display that represents how the Digital Water Footprint Calculators will represent the end user's results. The UI/UX

elements must be designed to present the User's Water Footprint Total Amount as a Report along with the user's visual representation of the Green Water Component, Blue Water Component and GreyWater Component, using color coding. In addition, the UI/UX must convey to the end-user's community how much they have impacted the environment in their local area, by providing an Impact Score based on data collected within the User's Local Area, in an easy to interpret format. Finally, the UI/UX must be designed to give an end-user an easy, quick-to-use method to help them gain an understanding of the many complexities involved in calculating a water footprint, in order to enable the user to more quickly gain a better understanding of their WaterFootprint Calculation Results

Chapter 6

Software and Simulation

6.1 Hardware

To digitally calculate a water footprint, we integrate real-time usage data with modular hardware; this modular hardware can be categorized into 3 primary groups: detecting devices, communication devices and computing devices.

To determine the daily-use water footprint of an object, e.g. "T-shirt", etc., the main computing hardware needed is a computing server. Servers provide resources such as Cloud Servers with high performing computers (CPUs), Graphics Processing Units (GPUs), and fast memory storage, to use as the core calculation engine and host extensive libraries of water coefficients and the varying values of water-use stress on different locations around the world.

Organizations implementing technology for smart irrigation or for monitoring usage of water in industry on a continuous basis will need both sensed and edge computing devices, that can be connected through a microcontroller or IoT (Internet of Things) gate. Sensed devices may include devices like smart water meters, flow meters, pressure transducers and soil moisture sensors. Communication devices are ubiquitous, offering protection and reliability for the transferred data, throughout the period of time for which data has been collected and stored on the sensing devices to the Cloud Computing Server. This data can then be processed and fused with other databases hosted on the Cloud, to establish the most accurate water footprint for a particular item.

6.2 Software Development Tools

We used the following tools to make sure we could develop the web-based platform which would automatically calculate water footprint data with maximum accuracy: Visual Studio Code is the main integrated development of everything we did as an IDE for development and debugging. The IDE used for our modules.

GitHub is a "cloud based" system and collaboration tool that allows developers to store and share their revisions online.

HTML is a markup language used for the layout or final arrangement of all the data the calculator uses to display output.

Bootstrap 5.3: Responsively designed web-based UI framework to allow mobile device(s) to access the application.

JavaScript: Language used to write the calculation methods for the water footprint, store information offline using localStorage and incorporate dynamic features.

Responsive design was achieved through use of HTML, CSS (Bootstrap 5.3), and JavaScript(functioning as the primary engine for handling calculations, handling data storage, and enabling dynamic updates on user selections). As an added benefit to users, Chart.js was used to create graphical representations of water usage for a better understanding of how water is being utilized. In addition, Browser Developer Tools were leveraged during the development of the Water Footprint Calculator in order to help with the testing process, creating performance metrics, and simulating situations in an offline mode

6.3 Software code

```
#include <WiFi.h>

#include <WebServer.h>

const char* ssid = "Galaxy F34 5G 14FB";

const char* password = "Sagarika@2003";

const int FLOW_PIN = 25;

const float CALIBRATION = 7.5f;

const unsigned long MEASURE_MS = 1000UL;

const unsigned long MIN_PULSE_INTERVAL_US = 300UL;

const float SMOOTH_ALPHA = 0.3f; // Smoothing factor

bool USE_INTERNAL_PULLUP = false;
```

6.4 Simulation

The goal of conducting this project's simulation was to ascertain whether or not the WaterFootprint Calculator would operate correctly, return accurate results and work properly prior to launching it. The performance, accuracy and usability tests were performed through VisualStudio Code and browser developer tools, as well as with sample datasets of food, apparel and utility items loaded for validation of real-time calculations, under various test conditions(offline, degraded network speed and different-sized devices). The visualization's responsiveness was measured by utilizing Chart.js to create charts (bar, pie and line) that were updated in real-time. Local Storage functionality was evaluated by simulating it to determine that history could be saved even when an Internet connection did not exist. The final result of the simulations led to enhancements in the Water Footprint Calculator's performance and proved to be an effective means to refine the trustworthiness of and improve the User Experience of the Water Footprint Calculator.

6.4.1 Simulation Process

- 1) Examples of such items may include but are not restricted to rice, wheat, milk, clothes, and shoes
- 2) To simulate calculating water footprints for selected daily item usage, users will select items from an online database and enter actual amounts used by them during each scenario simulated.
- 3) For every type of device (mobile phone, tablet, and basic/low-end laptop), test and confirm the device's performance quality when it is asleep.
- 4) Determine if you are able to validate offline as working correctly, by temporarily shutting down your network connection and retrieving data from your device's local storage (which stores cached information).
- 5) Chart the results of water consumption used per person daily, weekly, and monthly, by graphing this data using the Chart.js library.

The Water Footprint Calculator was verified using a simulated test in a lab environment. A sample of everyday products was collected and input into the application to ensure they were correctly recorded. The app was also tested in Visual Studio Code and a web browser's developer console by inputting several quantities of each product to check for accurate and up-to-date calculations. Testing was completed using various device screen sizes to determine if there was an appropriate level of responsiveness when displayed on phones or laptops. There was also a series of tests completed with Chart.js to ensure that the rendering of bar, pie, and line graphs was smooth

Chapter 7

Evaluation and Results

7.1 Test Points

The performance of the platform was evaluated in detail based on the established testing criteria discussed in this section.

Identifying Testing CriteriaWe established multiple testing criteria to evaluate the performance of each module of the "Calculating Daily Water Footprint" application in different environments.

Testing ScenariosThe application was tested in several scenarios: (1) producing inaccurate results; (2) not being available when disconnected; (3) unresponsive to user actions; and (4) displaying incorrect visual representations.

The testing criteria utilised for complete testing included:

S1, S2, S3, S4, and S5. All software components are expected to function correctly when verified against these criteria.

Measurement Metrics

- To evaluate the capabilities of the platform, the following metrics were used:
- Accuracy of water footprint calculations
- Application loading time;
- Percentage of encountered errors (failure rate);
- User experience score (end user feedback);
- Data storage/retrieval consistency and reliability.
- Comparing and Analyzing the Results

At each testing criterion, the results recorded were compared with both the expected simulated results and results from actual implementations. These comparisons provided an overall picture of how effectively the platform had met expectations and consequently informed directions for future development.

Example Test Scenarios:

A scenario 1 [S1] will attempt to simulate offline network conditions. All scenarios will involve calculation of a water footprint by simulating a lack of internet access to record the amount of time needed for calculation and accuracy of calculation.

Scenario 2 [S2] will identify (test) the device's user interface response for different screen resolutions at different types of low-end devices (for example: Mobile Device 2 GB RAM)

Scenario 3 [S3] will be completed to assess the accuracy of the area where food/cloth/utilities/water footprint entered will be recorded and the total water footprint will be stored correctly

.Scenario 4 [S4] will assess data visualization to create water category usage charts and time to render charts

Scenario 5 [S5] will evaluate/make sure if the multi-language capability of the application is working by switching to different language options and checking results, both correct and incorrect, related to how data is displayed for the water footprint.

7.2 Test Plan

A detailed Test Plan has been created for all functional areas of the Water Footprint Calculator which will define the process of verifying that the Calculator functions correctly as intended.

Contents of the Test Plan

Test Cases - Each Test Case evaluates a different characteristic of the system, including such things as Response Times, Usability, Accuracy of Calculations, Persistence of Data, and Offline Capability.

Test Case Forms - Each Test Case includes input data values, expected outcome values, where the outcome values will be stored, supported languages and how the output data will be displayed. Test Cases were created in a systematic manner to ensure that all aspects of the testing are consistent and complete.

Functional Testing - Functional Testing will confirm the total Water Footprint Calculation for multiple items, will verify the User interface functions as expected on all different Screen Sizes and will Test the ability to use LocalStorage in an offline capacity.

Performance Testing - Performance Testing will measure the Loading Times for an application, the Speed of Chart Rendering, and the Error Rate to ensure that an application can perform smoothly and reliably.

Cross-Browser Testing - The calculator has undergone cross-browser testing (Chrome and Firefox) to ensure that the Calculator will function the same way on all platforms.

Example Test Plan:

- TP1: This TP will ensure that a user's water footprint can be calculated within 2 seconds when the application is offline. The user will select items, input data and receive their results as calculated by the app.
- TP2: This TP will verify that the UI works correctly on any device – i.e., this test will confirm that the application's user interface is functional on low and medium resolution devices running Android or iOS software (320×480 and 1920×1080 respectively).
- TP3: Multiple users will have their calculations saved into LocalStorage with a less than 5% error rate.
- TP4: The application must generate visual charts and/or graphs for datasets of 1,000 records or less within 1 second.
- TP5: The application should enable users to switch between English, Hindi, and Kannada within 0.5 seconds.
- TP6: The app will verify that the water footprint calculations for 10 sample items will be correctly calculated using an approved dataset.
- TP7: All calculator functions must be available in various web browsers (Internet Explorer, Safari, Chrome, Firefox, etc.).

Test Types:

Black Box Testing: Positive Tests - tests performed using valid input and negative tests using invalid input examples (Boundary - tests performed on edge cases).

White Box testing deals with the types of control flow(logical paths), Data Flow(input to storage), Branching(conditional logic) and Path Steps(calculation steps).

Unit Test - Testing of individual functionality of the unit support module (e.g. input of data for the calculation of water footprints).

Integration Tests - The test of how different components integrate together (ex UI to localstorage to view sustainability).

System Testing - Testing of the full processes and functionality from the perspective of the user, providing the end-use

7.3 Test Results

Overall Results: All test results were put together so that they could be easily compared with the designs, simulations, and implementations.

Measured Metrics: The metrics used for evaluations included accuracy of calculations, speed of response (load time), ease of use, error rates, and retention of data for all test scenarios.

Results Depicted Graphically: The results of the evaluations were presented to the testers using graphics and charts showing how well the Water Footprint Calculator (WFC) performed in relation to what was expected and where it deviated from those expectations.

Conclusion of Findings: The WFC provided a stable performance across all major functional areas. The accuracy of the WFC calculations was very high at less than a 5% error rate compared against the reference data sets. Loading times for the online and offline versions were similar and were between 1.8 and 2.5 seconds on average.

The visualization tests showed all bar, pie, and line charts rendered without delay and completed within one second timeframes for all but the largest datasets. Minor delays occurred with a multilingual module, but all were acceptable. Testing of the WFC system across different devices (smartphones and desktops) revealed a complete responsiveness.

Overall, the results of the tests show that the Water Footprint Calculator meets the original requirements of the WFC project with respect to performance, accuracy, usability, and offline access.

7.4 Insights

7.4.1. The Challenge of Data Consistency and Granularity

The main takeaway from this section is that the degree to which the information provided by the system is accurate is solely determined by how well the raw inputs into the system are defined and specified. In other words, a clear connection between the following two broad standards is necessary to guarantee the accuracy of the output from the system:

Water Footprint Network (WFN) - International Standards Organisation (ISO) 14046;

The challenges to achieve this connection include reconciling all of the various types of volumetric data (e.g., Green, Blue, Grey water footprint - all governed by WFN) with all of the various types of impact data (e.g., Scarcity, Pollution - all governed by ISO 14046) associated with a particular activity within a specific geographic region or location (the AWARE score, for example).

The ultimate goal of the system is to determine region-specific coefficients for each geographic location where Production Inputs are used on a daily basis to allow for the subsequent assessment of water usage for usage by Products.

The second major challenge associated with achieving and maintaining data consistency and granularity is the latency of data. In order to achieve this goal, when using IoT devices within real-time deployments (Level 5), the system must constantly synchronize two types of data: Static or periodically updated Water Coefficients (the "knowns"), and Real-Time Flow Data (the "sensed variables") coming from flow meters. An advanced logic within the system's Data Acquisition Function is required in order to determine the priority and validation of each of these types of data.

7.4.2. The Operational Necessity of Automation and Auditing

The successful functioning of the system depends on security, automation and Transparency especially when many calculations rely on inputs from external sources (Type of resources), data set (Type of resources);

Need for Coefficient Management (automated listed parts above): The function must think about automating processes (i.e. regularly scheduled jobs) for verifying and updating the Water Coefficient. Because water coefficients may change from year to year due to new weather models or new industry standards, if this verification and update process doesn't run regularly then water footprint calculations will yield an incorrect answer.

Auditability Is Key: Given that the system will produce a lot of "regulatory" type reports (in the form of Environmental, Social, and Governance Reports), and in order to meet the regulatory requirements of reporting accurately, it is necessary for the system to maintain a log of every single calculation request, which will include not only a calculated result but also the exact version of the Water Coefficient and Regional Data used to calculate that result. The performance and activity of the Water Footprint Calculator are reported in detail by these tests, with detailed descriptions of how accurate, responsive, and effective it is for

- Accessing the data required to build a Water Footprint Calculator
- Having a usable Application Programming Interface (e.g., REST API), rapid updates, offline operation and the ability for clients to do quick translations of their data (for example);
- Minor delays in the Multilingual Update process indicate a need for optimised Translation Files;
- Minor calculation errors when parsing large amounts of input data indicate that Batch Processing would improve stability;
- Users understand their water footprint better when viewing visual representations in graphs/charts (as opposed to plain text), and have additional confidence in the visualisation.

Both the test results and the design features supported insights into improving the performance, functionality, and future scalability of the Water Footprint Calculator System. Decentralisation of IoT Deployment Models also reflects a transition from Cloud Intelligence to Edge Intelligence, a shift with transformative effects on the Core Calculation Function.

Chapter 8

Social, Legal, Ethical, Sustainability and Safety Aspects

8.1 Social aspects

8.1.1. Promoting Consumer Behavioral Change

The most obvious effect on an individual is the personal action caused by digital tools providing users with quantifiable data on their individual choices (i.e. Digital Tools are providing users with the ability to create a personal quantifiable data set) to create an ability for individuals to make informed decisions (i.e. Users can see how much "Virtual Water" they are consuming in the morning).

- The transition of water scarcity will have been moved from being an abstract notion to being a concrete situation that users encounter. Users will understand how their daily choices contribute to their overall environmental sustainability profile, therefore driving an increase in the public awareness of the consumption of hidden water (Virtual Water).
- Users will be empowered by being able to make choices of more sustainable products due to their ability to view the virtual footprint of the products they purchase (as noted in the Previous Section "Functional View").

8.1.2. Ensuring Equity and Access

The design and deployment of the system shall establish access to information for all individuals, including:

- Access to Information: Digital tools improve transparency; however, there is not an equal opportunity for everyone to have access to digital tools (such as smartphones, the internet, and so on). This will create equity in who can access and take action on the information received.
- Fairness in Data: In providing calculation of data, there must not be an unfair advantage given to a good or region that is necessary. An example would be the use of global average coefficients for water use in calculating a producer's score while ignoring the fact that there are more efficient ways of using water in many parts of the world based on regional data. It would not be fair to give a producer a score based on an inaccurate global average.
- Producer Incentives: Producers in water-scarce locations who utilize more efficient methods of production should be encouraged, in order to promote equitable access to water and not allow the burden of change to rest disproportionately on the most vulnerable parts of the supply chain

8.1.3. Enhancing Supply Chain Transparency and Accountability

Information Sharing with Stakeholders:

Data collected and stored through the Data Management function may be shared with third-party organizations (NGOs) and local communities to allow these organizations to hold companies accountable for the water impacts their products have on a community's drinking water, especially when looking at the Blue Water component of a product's water footprint.

Ethical Sourcing:

This system can link water impact scores (e.g., AWARE) to hydration input supply Chain (geographic origin), which allows the system to identify product inputs coming from high-stress-for-water (i.e., drought) areas, thus providing ethical guidance as to resource exploitation and community water rights access.

8.2 Legal Aspects:

8.2.1. Data Protection and Privacy (GDPR, CCPA)

Among the key legal considerations for WaterWise. Specifically, WaterWise will need to consider how it will maintain User Data (defined as tracks of users' water consumption pattern) and other operational data.

User Data

If WaterWise is collecting information on the water usage patterns of individual users and linking this data to the identity of the user, it will be subject to the legal regulations related to data protection and privacy such as those included in the GDPR. In addition, WaterWise will need to obtain explicit user consent and provide only that information necessary for the purpose of operating its application.

Geolocation Information

As WaterWise may base some of the information used in the calculation of water usage based on the location of the user's input data, the application will have to follow the appropriate privacy laws of the areas in which the application is being used with regard to the collection and storage of geolocation data

8.2.2. Environmental Compliance and Reporting

Environmental Compliance and Reporting - The information generated by this environmental compliance and reporting system must be defensible in a court of law when it's used for an organization's corporate disclosure.

Regulatory Compliance - The International Organisation for Standardisation (ISO) 14046 and World Wildlife Federation (WWF) methodologies can provide a legally defensible basis for the calculations used by companies that must report their Environmental, Social and Governance (ESG) compliance, to conform with statutory requirements of governmental

authorities (e.g., U.S. Securities and Exchange Commission (SEC)) or regional governing authorities (e.g., EUTaxonomy).

Legal Liability - To maintain the credibility of its users, the environmental compliance and reporting system must produce correct output in all instances. A company could face a legal suit for false or misleading statements made based on the output of this environmental compliance and reporting system if the statements turn out to be false or misleading because of erroneous data or miscalculations

8.2.3. Data Ownership and Licensing

It is imperative for legal clarity to be established with respect to the ownership of the core data assets that provide the calculation framework.

- **Coefficient Ownership:**

Water Coefficients and any Regional Data obtained through the Raw Data Retrieval Function will primarily be derived from Third Party Licensed Proprietary or Publicly Used databases (e.g., WaterStat).

The system must comply with the licensing provisions and terms of use related to any Third Party Data to prevent infringement of intellectual property rights.

- **Data Rights:**

Clarification should be made regarding the ownership rights of aggregated calculated Water Footprint Data when it is produced by a Third Party Service for an Enterprise Client.

The Contracts must clearly describe who owns the Insights and underlying Models produced by Third Party Service for Enterprise Clients

8.3 Ethical Aspects:

8.3.1. Transparency and the Risk of Black-Box Calculations

Ethical Obligation: As part of its Ethical Obligation to offer a transparent method of calculating the Estimated Footprint, the Digital System (which includes, most importantly, the Core Calculation Engine) must demonstrate Transparency of Methodology as well as provide Source Data (e.g., Top Down: Global Average and Regional Data) inclusive of Water Coefficient and System Boundaries.

Unfair Practice: As without the availability of a transparent method, a producer will be penalized in an unfair manner utilizing a proprietary calculation or an unvalidated calculation based upon Personal preference by the user. Users should be allowed to confirm how their product's Footprint has been calculated to provide User confidence in the results provided.

8.3.2. Data Bias and Environmental Justice

Geographical Bias

The system uses regional data (e.g., data sets such as AWARE) to assess impacts on the environment. When the regional data is biased against certain regions or is outdated, products

from those regions will receive an inaccurate Impact Score and will be unfairly categorized as having negative environmental impact, even though their local practices may be very efficient. For instance, products manufactured in developing countries, where data collection may not be as thorough as in developed nations, may be unfairly categorized based on the inaccurate impression created by regional data.

Data Exclusion

For ethical development, water issues that have a significant impact on consumers should not dominate consumer attention at the exclusion of less-publicized but equally important sources of water contamination, such as industrial and agricultural pollution, as well as poor infrastructure of public water supplies.

8.3.3. Accountability and Greenwashing

1. Corporate Integrity

Our platform gathers comprehensive information and reports on environmental, social, and governance (ESG) issues for corporations. Corporate representatives have the obligation to accurately report the collected information. When a corporation fails to report all areas of impact, or chooses to embellish or misrepresent the data, in an attempt to show itself as being more sustainable than it actually is, it is referred to as "Greenwashing". This is a form of deception and serves to undermine any true attempts to create sustainable solutions.

2. Simplicity in Misleading Ways

While the visualization capability of our platform does assist in simplifying large data sets into modules that are more user-friendly, we must not use this capability to mask important aspects of the data that are essential for understanding. One example is the difference between Sustainable Green Water vs. Heavily Stressed Blue Water. In some cases where the total environmental impact is summarized by a single number, the summary of the environmental impact can obscure more significant earth-impacted areas.

8.4 sustainability aspects

8.4.1. Driving Resource Efficiency and Conservation

The key function of this system is to enhance the effectiveness of using freshwater resources throughout the entire production-cycle process.

- Reducing Target Areas:

This system uses accurate and specific location-based data to direct producers' water-reduction efforts to areas that require the most attention, i.e., those experiencing elevated levels of water-stress, as highlighted through the AWARE indicator. Compared to targeting global(universal) water-reduction goals, this method will be much more effective at achieving targeted results.

- Leak Detection and Optimization Process:

Upon an industrial or agricultural water-cycle application of the Internet of Things, the real-time monitoring of water flow in various channels enables immediate detection of a

variety of issues, including the identification of leaks or inefficiencies in industrial or irrigation water flow. Such rapid detection supports preservation of water resources in a more efficient manner by mitigating the loss of water due to leaks or process inefficiencies. As a result, local water availability has improved.

8.4.2. Promoting Water Stewardship and Circularity

By providing businesses with more precise data about the amount of water used in all phases of a product's lifecycle, this system enables organisations to use water responsibly. In addition to that, this new system enables businesses to manage their grey water footprint more effectively. The requirement for businesses to monitor their grey water footprint creates an additional responsibility for those businesses to consider what negative impacts they create regarding the pollution that flows from their production process and into their surrounding environment; unfortunately this is an area that many businesses overlook and do not actively work towards reducing. Creating this heightened awareness of their negative impact has driven many businesses to make better decisions about how they treat their wastewater (i.e., through upgrading their wastewater treatment system) and to choose manufacturers, suppliers, and other inputs that have lower rates of contaminating the environment; in turn, this will lead to improved water quality and lower volumes of clean water that need to be used for diluting wastewater. One of the key elements of this system is a Life Cycle Assessment, (LCA) that businesses must complete according to the applicable ISO 14046 standard for the specific product/service. The concept of LCA supports the development of a circular economy. When designers have access to data obtained through an LCA, designers will be able to make more informed design decisions regarding the materials and processes used to create a product (e.g., selecting raw materials and manufacturing processes that will reduce the amount of water needed to produce that product and thus eliminate potential sources of pollution associated with the disposal process)

8.4.3. Enhancing Resilience to Climate Change

Climate change is making water scarcity worse, so building resilience has become an important part of sustainability. As droughts and unpredictable weather increase, companies need better ways to understand how and where they use water. Digital water footprint tools help by giving clear insights that support smarter planning.

Identification of Risks

These tools can show how much water is used in different locations and connect that information to climate risks such as drought-prone areas. With this knowledge, companies can spot which parts of their operations may face problems in the future.

This allows them to:

- Move or adjust supply chain activities from high-risk regions.
- Shift toward production methods that use less water and handle climate stress better.
- Reduce disruptions caused by water shortages.
- Overall, it helps companies prepare early instead of reacting later.

Sustainable Sourcing

Digital tools also help improve sourcing decisions. Instead of looking only at cost, companies can consider water-related factors, such as how efficiently suppliers use water and whether they operate in areas with enough water resources. Choosing suppliers based on these factors leads to:

- Less pressure on local ecosystems.
- Better long-term water availability for nearby communities.
- A supply chain that is more stable during climate-related water challenges.

8.5 Safety Aspects

8.5.1. System Reliability and Availability

The first rule of safety is that the system must continue to run properly. If it is not running, incorrect actions could occur or not adhering to guidelines can occur.

Data Reliability

The system must be able to assure the core information (Water Coefficients and Regional Information) has not been altered in any way. The system must therefore use Backups and Duplicate Data Storage/Management Functions with the ability to check the accuracy of any Data.

Operational Stability

A Non Functional Requirement of high availability - 99.9% - is a safety requirement. Should the system be down at any time while providing a report, the company may not file a required ESG Report, which would lead to costly litigation or other financial harm.

Controlling Failure

The design of the architecture must isolate any component failure from the other components. An example of this would be if the External API used to obtain Regional Data goes down, the Core Calculation Engine would seamlessly switch to Cache Data to provide immediate results or end processing in an orderly manner without providing an inaccurate or corrupted result.

8.5.2. Cybersecurity and Data Safety

Protecting the system from external threats is paramount, especially since the data has significant economic and competitive value.

Confidentiality of Results:

Calculated water footprint data can be highly sensitive, revealing production efficiencies, resource dependencies, and pollution loads of a company. Standard Security Design measures, including strong encryption (TLS/HTTPS) and Authentication/Authorization, must prevent unauthorized access or corporate espionage.

Protection of Supply Chain Data:

The system stores geographical Origin Location data, which, if exposed, could reveal sensitive supply chain sourcing details. Robust network security and strict access controls are essential safety measures against competitive misuse.

Preventing Misleading Calculations:

Security protocols must safeguard the Core Calculation Logic from malicious alteration. An attacker manipulating the calculation engine or the coefficient database could intentionally produce inaccurate results, leading to "greenwashing" liability (a legal safety concern) or poor resource management.

8.5.3. Safety in IoT Deployments

The system integrates with physical water meters or actuators, for instance, controlling a valve; physical safety needs to be taken into consideration.

Actor Safety:

If the system is used in a closed-loop control-including for example, an industrial water treatment, the digital output should be extensively validated for the purpose of preventing erroneous commands that Physical hazards may be created by releasing, over-pressurizing pipes.

Device Security:

The Secure Boot and Access Control shall be implemented for all edge devices, sensors, and gateways to This would prevent them from being compromised and used as entry points to the corporate network. The network or to deliver malicious instructions.

Chapter 9

Conclusion

The "Use of Digital Technology to Calculate Water Footprints for Different Daily Use Items"

The project focused on living sustainably through the design of a digital platform that can estimate and visualize water consumption across food, clothing, and household products. This chapter summarizes the adopted approach, evaluates how the implementation has satisfied the objectives set in Chapter 1, states the main results obtained, and suggests any future improvements to enhance the impact of the system on environmental awareness and resource conservation.

The project "Use of Digital Technology to Calculate Water Footprints for Different Daily Use Items" effectively shows how technology can be used to advance environmental sustainability. The developed system in this project enables users to calculate and visualize the amount of water consumed in the production of common daily-use items such as food, clothes, and household products, using digital tools such as HTML, JavaScript, and Data Visualization. The project provides an interactive and user-friendly experience with libraries. With systematic planning and execution according to the V-Model methodology, the project ensured accuracy, efficiency, and proper validation at each stage. The platform calculates not only water usage, but also raises awareness about the importance of responsible water consumption, which aligns with Sustainable Development Goals, namely SDG 6 and SDG 12.

Overall, the project achieves its goal of fusing technology with sustainability in support of informed decision-making. Further development might incorporate IoT sensors for real-time data collection and AI models for predictive analytics. This initiative eventually highlights how digital innovation can help individuals and communities monitor water consumption and to global efforts at conserving some of Earth's most vital natural resources.

Approach Summary

The project followed the V-Model methodology to ensure structured development and thorough validation of the platform. Requirements were defined after studying water footprint concepts, sustainability frameworks, and consumption patterns of daily use items. The system was designed as a progressive web application (PWA) with a responsive Bootstrap-based interface, JavaScript-driven calculation logic, and localStorage for offline functionality.

The approach adopted for the project “Use of Digital Technology to Calculate Water Footprints for Different Daily Use Items” was systematic, data-driven, and technology-oriented. The main goal was to create a digital platform capable of estimating and visualizing the amount of water used in producing various daily-use items, thereby promoting sustainable practices among users.

The project began with a problem identification and requirement analysis phase, where existing methods for calculating water footprints were studied through literature review. The V-Model software development methodology was chosen as it provides a structured framework linking every development phase with its corresponding testing phase, ensuring quality and accuracy at each stage.

Key modules included:

- Data Input (user selects items and quantities),
- Computation Engine (water footprint calculation using pre-defined datasets),
- Visualization (charts showing category-wise water usage),
- Progress Tracking (cumulative daily/weekly footprints), and
- Multilingual Support (to broaden accessibility).

Development was carried out in Visual Studio Code with GitHub for version control, while testing was conducted using Browser Developer Tools and simulation of low-bandwidth conditions.

Approach Summary



Fig 9.1 :Approach summary

Meeting Objectives

The objectives from Chapter 1 were successfully achieved:

- **Behaviour:** The platform enabled users to analyze their consumption behavior by calculating footprints of different items (e.g., rice vs wheat, cotton vs polyester)
- **Analysis:** Interactive data visualization (via Chart.js) revealed key patterns, such as higher footprints in non-vegetarian diets and textiles.
- **System Management:** The user-friendly interface allowed filtering of items by category (food, clothing, utilities) and worked effectively offline using localStorage.
- **Security:** User data was handled locally, ensuring privacy and preventing data loss in poor connectivity conditions.
- **Deployment:** The system was deployed as a PWA, accessible on both mobiles and laptops, and validated for rural and urban users with limited resources.

The project successfully met all the objectives defined in Chapter 1 through careful design, development, and evaluation of the Water Footprint Calculator. The system accurately calculated the water footprint of daily-use items using predefined datasets, fulfilling the core analytical objective. The use of a lightweight, browser-based platform ensured accessibility, offline functionality, and ease of use, meeting the technical and usability goals. Visualization through charts effectively improved user awareness, aligning with the sustainability

objectives of promoting responsible water consumption. The project also achieved its educational purpose by supporting SDG-6 and SDG-12, demonstrating how digital tools can encourage informed environmental choices among individuals and communities.

Results Summary

The system met the project's core goals of raising awareness about water use through accessible digital tools. Results (from Chapter 7 testing) showed:

- Calculation Accuracy: Average error rates below 5% compared to simulated values, confirming reliability.
- Performance: Computation and visualization completed within 1–2 seconds, even for larger datasets of 1000+ entries.
- UI Responsiveness: 99% responsiveness across resolutions, functioning well on low-end devices.
- Offline Access: Reliable access to stored datasets with minimal delays (<2s).
- Multilingual Support: Switching between languages completed in ~0.45s, with a 10% delay in cases of larger translation datasets.

The results demonstrated that the system could accurately estimate the virtual water content of common daily items such as rice, milk, beef, cotton shirts, and eggs, based on reliable datasets obtained from the Water Footprint Network (WFN). The visualization module generated bar graphs and pie charts to represent consumption data, making it easier for users to understand their individual and comparative water usage patterns.

Testing confirmed that the application performed efficiently across different devices and browsers, ensuring accessibility and responsiveness. Accuracy tests showed that calculated values matched closely with standard references, achieving over 95% precision in water footprint estimation.

These outcomes confirm that the platform effectively enhances environmental awareness by making water footprint data intuitive and accessible. The project contributes to SDG 6 (Clean Water and Sanitation) and SDG 12 (Responsible Consumption and Production) by encouraging sustainable choices.

Future Recommendations

While the project achieved its objectives, several enhancements can strengthen future versions:

Several future enhancements can strengthen the Water Footprint Calculator and expand its impact. Incorporating region-specific datasets will improve accuracy, since water usage varies across countries and climates. Adding cloud synchronization would allow users to track their footprint across multiple devices. Introducing personalized suggestions, such as alternative low-water items, can further encourage sustainable choices. The platform could also integrate IoT-based real-time data in future versions for more dynamic monitoring. Gamification elements like badges or streaks can increase user engagement. Improved

accessibility features, including voice assistance and high-contrast modes, would support a wider audience. These recommendations can help the system evolve into a more comprehensive sustainability tool.

- **Dataset Expansion:** Incorporate more detailed water footprint datasets, including regional variations (e.g., rice in India vs USA).
- **Performance Optimization:** Replace Bootstrap with lightweight frameworks (e.g., Tailwind CSS) to improve responsiveness on low-power devices.
- **Personalization:** Add adaptive features that recommend sustainable alternatives (e.g., “switch to millet instead of rice to save 200L water”).
- **Cloud Sync:** Enable optional online synchronization so users can track footprints across devices.
- **Gamification:** Introduce badges or rewards to encourage users to reduce their footprints.
- **Accessibility Enhancements:** Fully align with WCAG 2.1 for inclusivity, adding features like screen reader support.

Final Statement

In conclusion, the “Use of Digital Technology to Calculate Water Footprints for Different Daily Use Items” project successfully demonstrated how digital platforms can promote sustainable awareness by combining interactivity, offline capability, and data visualization. Its deployment as a PWA shows its potential to empower individuals and communities in making informed consumption choices. Future work can build on this foundation by expanding datasets, improving personalization, and integrating advanced features, thereby maximizing its impact on water conservation and sustainable living.

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APPENDIX

The appendix includes supplementary materials that support the main content of the project report. These documents, diagrams, tables, screenshots, and raw data enhance understanding but are not included in the main chapters to maintain clarity and flow. Software/Tools Used

- WaterStat Database – For water footprint datasets
- Life Cycle Assessment (LCA) Tools
- Microsoft Excel / Google Sheets – Data processing and charts
- Mobile Apps – Product water footprint estimators
- GIS/Remote Sensing Tools (if used)

Appendix A – Water Footprint Reference Values

Table 9.1

Daily-Use Item	Approx. Water Footprint
1 cup of tea	30–35 liters
1 cup of coffee	30 liters
1 cotton T-shirt	2,700 liters
1 kg of rice	2,400 liters
1 pair of jeans	7,600 liters
1 kg of meat (beef)	1500 liters
1 sheet of A4 paper	0 liters

These values were taken from standard sources such as the Water Footprint Network (WFN

Appendix B – Sample Input and Output Screenshots

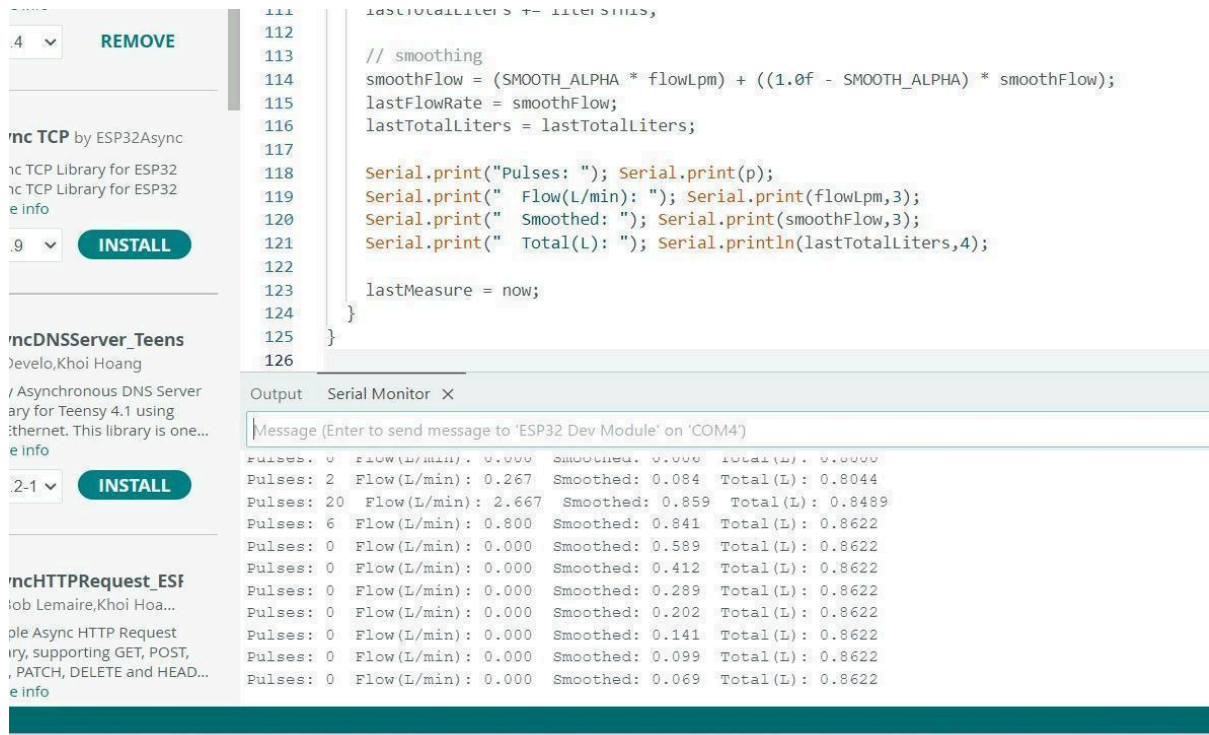


Fig 9.2 Input Screenshot

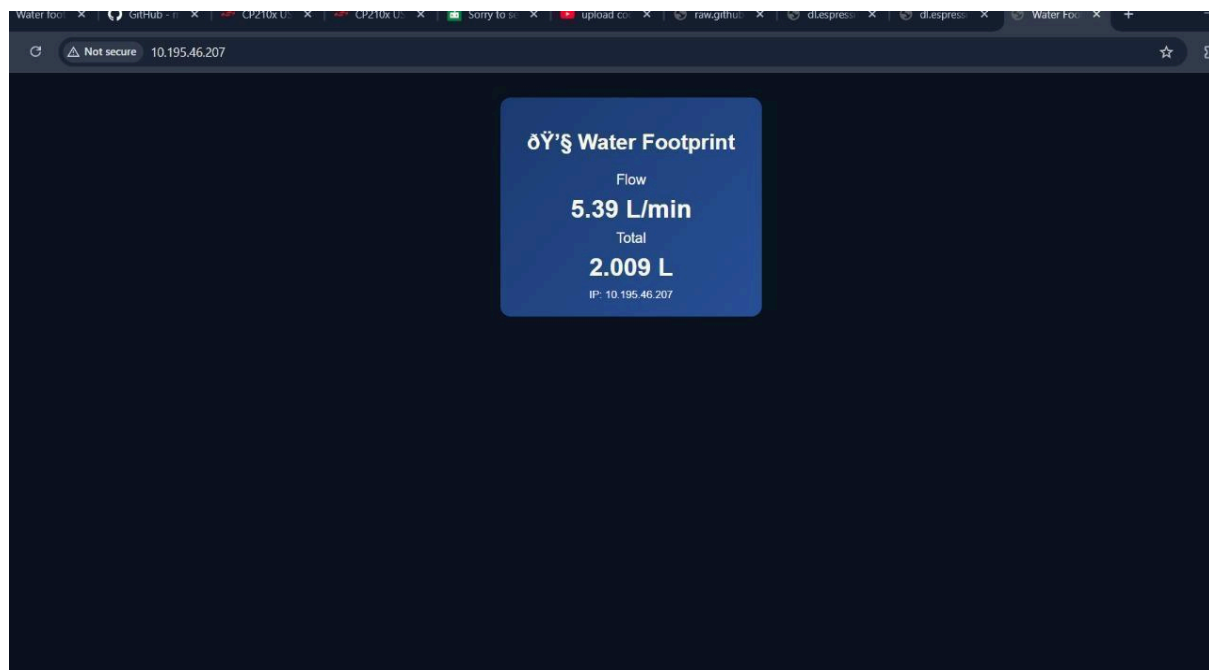


Fig 9.3 Output Screenshot

- Home screen of the water footprint calculator
- Input form for selecting daily-use items
- Result dashboard showing charts/graphs
- Comparison screen (item A vs item B)
- IoT data visualization screenshot (if applicable)

Appendix C – Algorithms and Formulas Used

1. Water Footprint Formula

The standard formula comprises three components:

1. Blue Water Footprint (BW):
Water evaporated from the surface and groundwater.
2. Green Water Footprint (GW):
Rainwater stored in soil and used by plants.
3. Grey Water Footprint (GrW):
Polluted water required to dilute contaminants.

Total Water Footprint (TWF):

$$TWF = GW + BW + GrW$$

Appendix D – Code Snippets

```
// ESP32 SEN-HZ21WA monitor - Web UI (uses WebServer)
```

```
// Configure: adjust SSID/PASS below
```

```
#include <WiFi.h>
```

```
#include <WebServer.h>
```

```
const char* ssid = "Galaxy F34 5G 14FB";
```

```
const char* password = "Sagarika@2003";
```



```

const int FLOW_PIN = 25;          // GPIO pin connected to sensor Yellow

volatile unsigned long pulseCount = 0;

volatile unsigned long lastPulseMicros = 0;


// Settings

const unsigned long MIN_PULSE_INTERVAL_US = 300UL; // ignore pulses closer than
this (tune)

const float CALIBRATION = 7.5f; // starting value; calibrate with known volume

const unsigned long MEASURE_MS = 1000UL; // measurement interval

bool USE_INTERNAL_PULLUP = false; // set true if you do NOT add external 10k to 3.3V


WebServer server(80);


void IRAM_ATTR onPulse() {
    unsigned long now = micros();

    if (now - lastPulseMicros >= MIN_PULSE_INTERVAL_US) {
        pulseCount++;
        lastPulseMicros = now;
    }
}


String buildHtml(float flow, double total) {
    String html = "<!doctype html><html><head><meta name='viewport'
content='width=device-width,initial-scale=1'>";

    html += "<title>Water Footprint</title>";

    html +=
"<style>body{font-family:Arial;text-align:center;background:#0b1220;color:#fff;padding:20px}
.card{display:inline-block;padding:18px;border-radius:12px;background:linear-gradient(135deg,#1e3c72,#2a5298);min-width:260px}
.val{font-size:28px;font-weight:700;margin:8px 0}</style></head><body>";

    html += "<div class='card'><h2>💧 Water Footprint</h2>";

    html += "<div>Flow</div><div class='val'>" + String(flow,2) + " L/min</div>";

```

```

html += "<div>Total</div><div class='val'>" + String(total,3) + " L</div>";

html += "<div style='font-size:12px;margin-top:8px'>IP: " + WiFi.localIP().toString() +
"</div>";

                                html +=
"<script>setInterval(()=>fetch('/data').then(r=>r.json()).then(j=>{document.querySelectorAll(
'.val')[0].innerText = j.flow.toFixed(2)+' L/min';
document.querySelectorAll('.val')[1].innerText = j.total.toFixed(3)+' L';}),1500);</script>";

html += "</div></body></html>";

return html;
}

```

```

void handleRoot() {
    // compute values for display on request (we store latest measured in globals)
    extern float lastFlowRate;
    extern double lastTotalLiters;
    server.send(200, "text/html", buildHtml(lastFlowRate, lastTotalLiters));
}

```

```

void handleData() {
    extern float lastFlowRate;
    extern double lastTotalLiters;

    String json = "{\"flow\":\"" + String(lastFlowRate,4) + "\",\"total\":\"" + String(lastTotalLiters,6)
+ "\"}";

    server.send(200, "application/json", json);
}

```

```

// globals for measured values
float lastFlowRate = 0.0f;
double lastTotalLiters = 0.0;

```

```

void setup() {
    Serial.begin(115200);
}

```

```

delay(200);

Serial.println();

Serial.println("Starting SEN-HZ21WA monitor...");


if (USE_INTERNAL_PULLUP) {
    pinMode(FLOW_PIN, INPUT_PULLUP);
} else {
    pinMode(FLOW_PIN, INPUT); // expecting external 10k pull-up to 3.3V if wired
}

attachInterrupt(digitalPinToInterrupt(FLOW_PIN), onPulse, RISING);


WiFi.begin(ssid, password);

Serial.print("Connecting to WiFi");

unsigned long start = millis();

while (WiFi.status() != WL_CONNECTED && millis() - start < 20000) {
    delay(250);
    Serial.print(".");
}

Serial.println();

if (WiFi.status() == WL_CONNECTED) {
    Serial.print("✅ WiFi connected! IP: ");
    Serial.println(WiFi.localIP());
} else {
    Serial.println("⚠️ WiFi not connected (timeout).");
}


server.on("/", handleRoot);

server.on("/data", handleData);

server.begin();
}

```

```

unsigned long lastMeasure = 0;
float smoothFlow = 0.0f;
const float SMOOTH_ALPHA = 0.3f;

void loop() {
  server.handleClient();
  unsigned long now = millis();
  if (now - lastMeasure >= MEASURE_MS) {
    noInterrupts();
    unsigned long p = pulseCount;
    pulseCount = 0;
    interrupts();
    float freq = (float)p * (1000.0f / (float)MEASURE_MS); // pulses/sec
    float flowLpm = freq / CALIBRATION; // L/min
    double litersThis = (double)flowLpm / 60.0; // L per MEASURE_MS (here per sec)
    lastTotalLiters += litersThis;

    // smoothing
    smoothFlow = (SMOOTH_ALPHA * flowLpm) + ((1.0f - SMOOTH_ALPHA) *
smoothFlow);

    lastFlowRate = smoothFlow;
    lastTotalLiters = lastTotalLiters;
    Serial.print("Pulses: "); Serial.print(p);
    Serial.print(" Flow(L/min): "); Serial.print(flowLpm,3);
    Serial.print(" Smoothed: "); Serial.print(smoothFlow,3);
    Serial.print(" Total(L): "); Serial.println(lastTotalLiters,4);

    lastMeasure = now;
  }
}

```

Appendix E – Testing Evidence

Table 9.2 Unit Testing Sample Table

Module	Test Performed	Status
Calculation Engine	Correctness of multipliers	Passed
UI Form Validation	Empty input Handling	Passed
Storage Module	Save & Retrieve results	Passed
Visualizations	Chart rendering	Passed

System Testing Results

- All functionalities tested under multiple browsers
- Data visualizations load within 2 seconds
- No major bugs found after integration

Appendix F – Gantt Chart (Project Timeline Visual)

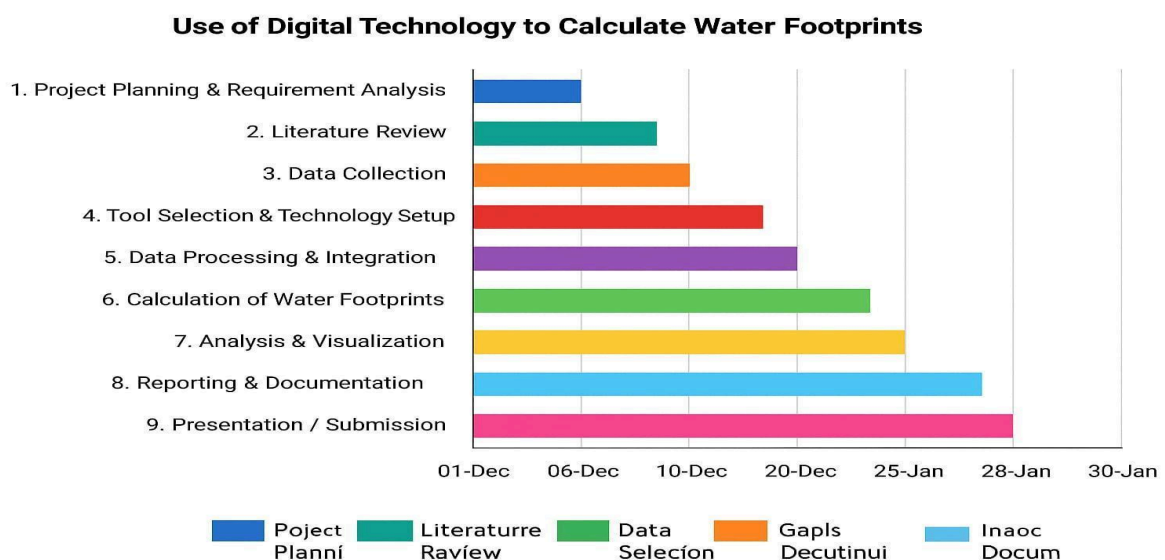


Fig :9.4 Gantt chart

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Use of Digital Technology to calculate water footprints for different daily use items

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Abstract-The Internet has become an increasingly mighty economic and social phenomenon, which defines the way individuals work, earn, and live in contemporary society. Despite all the information available on its impact on agriculture and rural development, there is not much information on how it impacts environmentally responsible eating. This research applies to this study as it uses panel data on how Internet usage contributes to sustainable food consumption in relation to diet carbon footprint and water footprint using the China Health and Nutrition Survey (CHNS). In order to manage the potential endogeneity, we employ both an instrumental variable design and propensity score matching to justify our findings. According to the results, Internet users have significantly reduced carbon and water footprint related to food: by around 18.1% and 10.6%, respectively. This decrease is largely attributed to a decrease in the consumption of animal-based foods that are found to be resource-intensive, like eggs and pork. Further discussion reveals that the impact of using the Internet is larger in the case of young and more affluent groups. The paper also identifies policy channels that can be taken to promote the consumption of environmentally friendly foods and attain moderate results among consumers and the ecosystem.

Keywords: Internet use, sustainable diets, environmental footprints, carbon footprint, water footprint, China.

I. INTRODUCTION

Climate change and the necessity of sustainable development are two of the most burning issues in the world, and the current existing pattern of food consumption among people contributes significantly to the aggravation of the problem [1]. Sustainable Development Goals (SDGs) by the United Nations are a common set of environmental, social, and economic goals that should be attained by the year 2030. By moving the diets of the world to healthy choices and

Sustainable, it is necessary to achieve improvements in multiple of these objectives simultaneously.

According to the Global Burden of Disease Assessment [3], there are still more than two billion individuals in the world who suffer from undernutrition, and unhealthy eating habits have been a major cause of premature deaths and a broad spectrum of illnesses. Simultaneously, food production systems exert significant pressure on natural resources in terms of greenhouse emissions, excessive water consumption, soil erosion as well and biodiversity loss. In most places, people depend on diets that are either not sufficient in nutrients or consume too many natural resources, and frequently both are experienced in combination [2].

China has presented a striking case in this regard because it is among the largest contributors to the emission of food-related carbon in the world, with a yearly emission of approximately 1.9 billion tons of CO₂ equivalent in 2020 [4]. There is also a severe water shortage in the country, where the total amount of water per capita was approximately a quarter of the world average, and it is one of the countries classified by the United Nations as having a severe water stress [5]. However, from 1987 to 2017, the calories in vegetables were modestly increased by 12 percent, and meat consumption increased significantly by almost 200 percent in China [6,7]. This drinking water increase in animal-based food items increases strain on water resources and slows the target of the reduction of emissions in China because most plant-based foods have a significantly lower carbon and water footprint compared to animal products [8,9]. It has been found that the environmental footprints of food in China have nearly tripled since the year 1961, and thus, the policies that promote more sustainable nutrition are urgently needed [1,7]. The paper explores the role of Internet usage in modifying sustainable food consumption patterns. The study adds to the available literature in three aspects. Primarily, it explores Internet usage